Leverage, Structural Rank, and Thermodynamic Selection in Information-Processing Systems

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Abstract

We study leverage as a structural invariant of information-processing systems. For any system A with capabilities $\mathrm{Cap}(A) > 0$, we prove a Five-Way Equivalence:

$$\begin{aligned} \mathrm{DOF}(A) = 1 &\iff \text{max leverage } L = |\mathrm{Cap}|/\mathrm{DOF} \iff \text{single source of truth} \\ &\iff \text{srank} = 1 \iff \text{tractable sufficiency} \\ &\iff \text{minimum thermodynamic cost.} \end{aligned}$$

We also prove that the England replication inequality,

$$\Delta S_{\min}(k) - \Delta S_{\min}(1) \geq k_B \ln k,$$

reduces to counting: a k-variable system has 2^k states, and the finite inequality $k \leq 2^{k-1}$ yields the entropy gap under Landauer calibration. The thermodynamic bound $E_{\text{decision}} \geq \text{srank} \cdot k_B T \ln 2$ then follows from the same calibration (BA7).

Within the explicit thermodynamic model used in the paper (Landauer calibration, finite energy, finite signal speed, and nontrivial state space), we derive a physical no-go theorem for polynomial collapse (LP38). Engineering corollaries follow directly: $\mathrm{DOF} = n$ is isomorphic to a series reliability system with n failure points; expected modification cost is proportional to DOF; and minimum edit-energy equals $k_B T \ln 2 \cdot \mathrm{DOF}(A)$.

All theorems are machine-checked in Lean 4 with no `sorry` placeholders. The only explicit physics premises are Landauer calibration, the Second Law (non-negative entropy production), and the Thermodynamic Uncertainty Relation (Barato-Seifert 2015). An assumption ledger records the dependency structure of each physically grounded result.

Keywords: thermodynamics, Landauer principle, entropy production, information theory, structural rank, formal verification, degrees of freedom

1 Introduction

This paper proves thermodynamic bounds on information-processing systems from first principles. The central result is the Five-Way Equivalence (Theorem 1.1): five independent scientific frameworks (engineering, epistemics, information theory, computational complexity, and statistical physics) all characterize the same structural property, $\mathrm{DOF} = 1$. Software architecture serves as the application domain; the underlying physics is general. All claims are verified in Lean 4 with zero `sorry` placeholders.

1.1 Formalization Statistics

Metric	Value
Lines of Lean 4 (local layer)	4120
Theorems/lemmas (local layer)	210
`sorry` placeholders (local layer)	0
Proof files (local layer)	17
Total imported Lean lines	48509
Total theorem/lemma statements	2079
Total imported `sorry` placeholders	0
Total proof files in import closure	195
Dependencies (live imports)	
`AbstractClassSystem`	7215 lines, 306 theorems
`Ssot`	3863 lines, 185 theorems
`DecisionQuotient`	33311 lines, 1378 theorems

All proofs build with `lake build`. Axiom dependencies verified via `#print axioms`: `propext`, `Quot.sound`, `Classical.choice`.

1.2 Central Result

Theorem 1.1 (Five-Way Equivalence)**.** *For any architecture A with $\mathrm{Cap}(A) > 0$, the following are equivalent:*

$$\underbrace{\mathrm{DOF}(A) = 1}_{\textit{single source}} \iff \underbrace{L(A) \textit{ maximal}}_{\textit{engineering}} \iff \underbrace{\mathrm{SSOT}(A)}_{\textit{epistemics}}$$
$$\iff \underbrace{\mathrm{srank}(A) = 1}_{\textit{information}} \iff \underbrace{\textit{tractable sufficiency}}_{\textit{complexity}} \iff \underbrace{\textit{min thermo cost}}_{\textit{physics}}$$

Proved in Section 5. Machine-checked via `Leverage` → `DecisionQuotient` → `Ssot` → Mathlib.

1.3 Contributions

1. **DOF-Reliability Isomorphism (Theorem 3.9):** Architecture with n DOF is isomorphic to a series reliability system with n components.

2. **Leverage-Error Tradeoff (Theorem 4.2):** $L(A_1) > L(A_2) \implies P_{\text{error}}(A_1) < P_{\text{error}}(A_2)$ at equal capabilities.

3. **Modification Complexity Gap (Theorem 3.11):** Expected modification cost is proportional to DOF at fixed capabilities.

4. **Physical Edit-Energy Floor (Theorem 4.3):** Minimum edit-energy $= j_{\mathcal{M}} \cdot \mathrm{DOF}(A)$ under Landauer calibration. No P $\neq$ coNP assumption. *(L: L55, L49)*

5. **England Replication Inequality (Theorem 5.1):** $\Delta S_{\min}(k) - \Delta S_{\min}(1) \geq k_B \ln k$. Proof: a k-variable system has 2^k states (counting); entropy gap follows from $k \leq 2^{k-1}$ (induction). *(L: L45)*

6. **Optimal Architecture (Theorem 4.8):** $f(R) = \arg\max_{A:\, \mathrm{Cap}(A) \supseteq R} L(A)$ minimizes expected error and is computable.

What is new. `AbstractClassSystem` and `Ssot` establish two of the five equivalences. This paper adds leverage maximization and proves it equivalent to the remaining three. The England inequality is new: the only physics input is the Landauer constant k_B; the bound reduces to $k \leq 2^{k-1}$.

1.4 Dependency Chain

1. `AbstractClassSystem`: axis orthogonality $\to$ error independence (Theorem 3.1)
2. `Ssot`: DOF $= 1 \iff$ coherence $\to$ second equivalence
3. `DecisionQuotient`: tractability boundary and thermodynamic lift $\to$ fourth and fifth equivalences
4. `Leverage` (this paper): leverage maximization $\iff$ all of the above

1.5 Scope

Leverage measures capability-to-DOF ratio. Performance, security, and other dimensions are orthogonal. Error independence is derived from axis orthogonality in `AbstractClassSystem`, not assumed. Physical claims use one explicit constant ($j_{\mathcal{M}} > 0$) and no additional axioms beyond the Landauer calibration.

1.6 Organization

Section 2 defines Architecture, DOF, Capabilities, and Leverage. Section 3 derives the error model. Section 4 proves decidability and optimality. Section 6 gives leverage instances. Section 5 proves the five-way equivalence and the England inequality. Section A describes the Lean mechanization.

2 Foundations

We formalize the core concepts: architecture state spaces, degrees of freedom, capabilities, and leverage.

2.1 Architecture State Space

Definition 2.1 (Architecture)**.** An *architecture* is a tuple $A = (C, S, T, R)$ where:

- C is a finite set of *components* (modules, services, endpoints, etc.)
- $S = \prod_{c \in C} S_c$ is the *state space* (product of component state spaces)
- $T : S \to \mathcal{P}(S)$ defines valid *transitions* (state changes)
- R is a set of *requirements* the architecture must satisfy

Intuition: An architecture consists of components, each with a state space. The total state space is the product of component spaces. Transitions define how the system can evolve.

Example 2.2 (Microservices Architecture)**.**

- $C = \{U, O, P\}$ where $U =$ UserService, $O =$ OrderService, $P =$ PaymentService
- $S_{\text{UserService}} = \text{UserDB} \times \text{Endpoints} \times \text{Config}$

- Similar for other services
- $S = S_{\text{UserService}} \times S_{\text{OrderService}} \times S_{\text{PaymentService}}$

2.2 Degrees of Freedom

Definition 2.3 (Degrees of Freedom)**.** The *degrees of freedom* of architecture $A = (C, S, T, R)$ is a natural number $\text{DOF}(A) \in \mathbb{N}$ counting the independent axes of variation in the state space. Formally, it is the cardinality of a minimal complete orthogonal axis set for S (as established in `AbstractClassSystem`). In the Lean formalization, `Architecture.dof` is a field of type ℕ.

Operational meaning: DOF counts independent modification points. If $\text{DOF}(A) = n$, then n independent changes can be made to the architecture.

Proposition 2.4 (DOF Additivity)**.** *For disjoint architectures $A_1 = (C_1, S_1, T_1, R_1)$ and $A_2 = (C_2, S_2, T_2, R_2)$ with $C_1 \cap C_2 = \emptyset$:*

$$DOF(A_1 \oplus A_2) = DOF(A_1) + DOF(A_2)$$

where $A_1 \oplus A_2 = (C_1 \cup C_2, S_1 \times S_2, T_1 \times T_2, R_1 \cup R_2)$.

(L: `L17`, `L19`)

Proof. Components are disjoint ($C_1 \cap C_2 = \emptyset$), so the axis sets of A_1 and A_2 are disjoint. The axis set of $A_1 \oplus A_2$ is their disjoint union, giving $\text{DOF}(A_1 \oplus A_2) = |\text{axes}(A_1)| + |\text{axes}(A_2)| = \text{DOF}(A_1) + \text{DOF}(A_2)$. ■

Example 2.5 (DOF Calculations)**.**

1. **Monolith:** Single deployment unit → DOF = 1
2. n **Microservices:** n independent services → DOF $= n$
3. **Copied Code:** Code duplicated to n locations → DOF $= n$ (each copy independent)
4. **SSOT:** Single source, n derived uses → DOF = 1 (only source is independent)
5. k **API Endpoints:** k independent definitions → DOF $= k$
6. m **Config Parameters:** m independent settings → DOF $= m$

2.3 Capabilities

Definition 2.6 (Capability Set)**.** The *capability set* of architecture A is:

$$\text{Cap}(A) = \{r \in R \mid A \text{ satisfies } r\}$$

Examples of capabilities:

- "Support horizontal scaling"
- "Provide type provenance"
- "Enable independent deployment"
- "Satisfy single source of truth for class definitions"
- "Allow polyglot persistence"

Definition 2.7 (Capability Satisfaction)**.** Architecture A *satisfies* requirement r (written $A \vDash r$) if there exists an execution trace in (S, T) that meets r's specification.

2.4 Leverage

Definition 2.8 (Leverage)**.** The *leverage* of architecture A is:

$$L(A) = \frac{|\text{Cap}(A)|}{\text{DOF}(A)}$$

Special cases:

1. **Unbounded Leverage:** As $|\text{Cap}(A)|$ grows for fixed $\text{DOF}(A) = 1$, leverage grows without bound. This is the metaprogramming regime: a single source with an unbounded number of derived capabilities. In any fixed finite model, L is a positive rational; "$L = \infty$" is a shorthand for this limit.

2. **Unit Leverage ($L = 1$):** Linear relationship (n capabilities from n DOF)

3. **Sublinear Leverage ($L < 1$):** Antipattern (more DOF than capabilities)

Example 2.9 (Leverage Calculations)**.**

- **SSOT:** DOF $= 1$, Cap $= \{F, \text{uses of } F\}$ where each new use adds a capability
  $\Rightarrow L$ grows without bound as derivations accumulate

- **Scattered Code (n copies):** DOF $= n$, Cap $= \{F\}$
  $\Rightarrow L = 1/n$ (antipattern!)

- **Generic REST Endpoint:** DOF $= 1$, Cap $= \{\text{serve } n \text{ use cases}\}$
  $\Rightarrow L = n$

- **Specific Endpoints:** DOF $= n$, Cap $= \{\text{serve } n \text{ use cases}\}$
  $\Rightarrow L = 1$

Definition 2.10 (Architectural Dominance)**.** Architecture A_1 *dominates* A_2 (written $A_1 \succeq A_2$) if:

1. $\text{Cap}(A_1) \supseteq \text{Cap}(A_2)$ (at least same capabilities)

2. $L(A_1) \geq L(A_2)$ (at least same leverage)

A_1 *strictly dominates* A_2 (written $A_1 \succ A_2$) if $A_1 \succeq A_2$ with at least one inequality strict.

2.5 Modification Complexity

Definition 2.11 (Modification Complexity)**.** For requirement change δR, the *modification complexity* is:

$$M(A, \delta R) = \text{expected number of independent changes to implement } \delta R$$

Theorem 2.12 (Modification Bounded by DOF)**.** *For all architectures A and requirement changes δR:*

$$M(A, \delta R) \leq DOF(A)$$

with equality when δR affects all components.

(L: L37)

Proof. By Definition 2.3, DOF counts independent axes. A requirement change δR decomposes into sub-changes along individual axes; sub-changes along distinct axes are independent. Therefore the number of independent changes is at most the number of axes, which is $\text{DOF}(A)$. Equality holds when δR has a non-trivial projection onto every axis. ■

Example 2.13 (SSOT vs Scattered)**.** Consider changing a structural fact F with n use sites:

- **SSOT:** $M = 1$ (change at source, derivations update automatically)
- **Scattered:** $M = n$ (must change each copy independently)

2.6 Formalization in Lean

All definitions in this section are formalized in `Leverage/Foundations.lean`:

- `Architecture`: Structure with components, state, transitions, requirements
- `Architecture.leverage`: Leverage metric
- `Architecture.dominates`: Dominance relation
- `compose_dof`: Proposition 2.4
- `modification_eq_dof`: Theorem 2.12

3 Probability Model

We derive the relationship between DOF and error probability from `AbstractClassSystem`'s axis independence theorem.

3.1 Error Independence from Axis Orthogonality

The independence of errors is not an axiom; it is a consequence of axis orthogonality proved in `AbstractClassSystem` [24].

Theorem 3.1 (Error Independence)**.** *If axes $\{A_1, \ldots, A_n\}$ are orthogonal (`AbstractClassSystem`, Theorem `minimal_complete_unique_orthogonal`), then errors along each axis are statistically independent.*

(L: `L23`)

Proof. By Definition 2.1, the state space is the product $S = \prod_{c \in C} S_c$. `AbstractClassSystem`'s orthogonality theorem (`minimal_complete_unique_orthogonal`) establishes that a minimal complete axis set partitions the state space into independent dimensions: axis A_i is a projection onto a factor of the product, and $A_i \perp A_j$ means the projection onto factor i carries no information about factor j.

An *error along axis* A_i is the event E_i that the value on factor i deviates from specification. Since the factors are independent dimensions of the product space, the probability measure factorizes:

$$P(E_i \mid \text{state of factors } \neq i) = P(E_i).$$

This is the standard characterization of independence for product probability spaces: the marginal on factor i is independent of the marginal on any other factor. Therefore $P(E_i \cap E_j) = P(E_i) \cdot P(E_j)$ for all $i \neq j$, and by induction the n events $E_1, \ldots, E_n$ are mutually independent. ■

Corollary 3.2 (DOF = Independent Error Sources). *$DOF(A) = n$ implies n independent sources of error, each with probability p.*

(L: L15 , L23)

Proof. DOF counts independent axes (Section 2, Definition 2.3). By Theorem 3.1, independent axes have independent errors. ■

Theorem 3.3 (Error Compounding). *For a system to be correct, all n independent axes must be error-free. Errors compound multiplicatively.*

(L: L41 , L20)

Proof. By `Ssot`'s coherence theorem (`oracle_arbitrary`), incoherence in any axis violates system correctness. An error in axis A_i introduces incoherence along A_i. Therefore, correctness requires $\bigwedge_{i=1}^{n} \neg\text{error}(A_i)$. By Theorem 3.1, this probability is $(1-p)^n$. ■

3.2 Error Probability Formula

Theorem 3.4 (Error Probability). *For architecture with n DOF and per-component error rate p:*

$$P_{error}(n) = 1 - (1-p)^n$$

(L: L40 , L24)

Proof. By Theorem 3.1 (derived from `AbstractClassSystem`'s orthogonality), each DOF has independent error probability p, so each is correct with probability $(1-p)$. By Theorem 3.3, all n DOF must be correct:

$$P_{\text{correct}}(n) = (1-p)^n$$

Therefore:

$$P_{\text{error}}(n) = 1 - P_{\text{correct}}(n) = 1 - (1-p)^n$$

■

Corollary 3.5 (Linear Approximation). *For fixed $p > 0$, the linear expected-error model and the exact series model induce the same ordering on architectures by DOF.*

(L: L39 , L16)

Proof. By Theorem 3.10, ordering equivalence is exact under the discrete model used in the mechanization. ■

Corollary 3.6 (DOF-Error Monotonicity). *For architectures A_1, A_2:*

$$DOF(A_1) < DOF(A_2) \implies P_{error}(A_1) < P_{error}(A_2)$$

(L: L33)

Proof. $P_{\text{error}}(n) = 1 - (1-p)^n$ is strictly increasing in n for $p \in (0,1)$. ■

3.3 Expected Errors

Theorem 3.7 (Expected Error Bound)**.** *Expected number of errors in architecture A:*

$$\mathbb{E}[\#\ errors] = p \cdot DOF(A)$$

(L: L25 , L26)

Proof. By linearity of expectation:

$$\mathbb{E}[\#\ \text{errors}] = \sum_{i=1}^{\text{DOF}(A)} P(\text{error in DOF}_i) = \sum_{i=1}^{\text{DOF}(A)} p = p \cdot \text{DOF}(A)$$

■

Example 3.8 (Concrete Calculations)**.** Assume $p = 0.01$ (1% per-component error rate):

- DOF $= 1$: $P_{\text{error}} = 1 - 0.99 = 0.01$ (1%)
- DOF $= 10$: $P_{\text{error}} = 1 - 0.99^{10} \approx 0.096$ (9.6%)
- DOF $= 100$: $P_{\text{error}} = 1 - 0.99^{100} \approx 0.634$ (63.4%)

3.4 Connection to Reliability Theory

The error model has a direct interpretation in classical reliability theory [20], connecting software architecture to a mature mathematical framework with 60+ years of theoretical development.

Theorem 3.9 (DOF-Reliability Isomorphism)**.** *An architecture with $DOF = n$ is* isomorphic *to a series reliability system with n components. The isomorphism:*

1. ***Preserves ordering:*** *If $DOF(A_1) < DOF(A_2)$, then $P_{error}(A_1) < P_{error}(A_2)$*
2. ***Is invertible:*** *Round-trip mapping preserves DOF exactly*
3. ***Connects domains:*** *$P_{error}(n) = 1 - R_{series}(n)$ where $R_{series}(n) = (1 - p)^n$*

(L: L22 , L27)

Interpretation: Each DOF is a "component" that must work correctly. This is the reliability analog of Theorem 3.1, which derives error independence from axis orthogonality.

Theorem 3.10 (Ordering Equivalence (Exact))**.** *For architectures A_1, A_2 with equal capabilities and error rate $p > 0$, the linear model and the exact series model induce the same DOF ordering:*

$$DOF(A_1) \leq DOF(A_2) \iff (\mathbb{E}[errors(A_1)] \cdot d) \leq (\mathbb{E}[errors(A_2)] \cdot d),$$

where d is the denominator of the discrete error-rate representation.

(L: L39 , L32)

Proof. This is theorem `ordering_equivalence_exact` in `Leverage/Probability.lean`. ■

Linear Approximation Justification: For small p (the software engineering regime where $p \approx 0.01$), the linear model $P_{\text{error}} \approx n \cdot p$ is:

1. Order-equivalent to the exact series model in the mechanized discrete representation
2. Monotone in DOF for fixed positive error rate
3. Cleanly formalized in natural-number arithmetic

3.5 Epistemic Grounding

The probability model is not axiomatic; it is derived from the epistemic foundations in `AbstractClassSystem` and `Ssot`:

1. `AbstractClassSystem` proves axis orthogonality (`minimal_complete_unique_orthogonal`)
2. **Theorem 3.1** derives error independence from orthogonality
3. `Ssot` establishes that DOF = 1 guarantees coherence (`oracle_arbitrary`)
4. **Theorem 3.3** connects errors to incoherence

This derivation chain ensures the probability model rests on proved foundations, not assumed axioms.

3.6 Leverage Gap: Quantitative Predictions

The leverage framework provides *quantitative*, empirically testable predictions about modification costs.

Theorem 3.11 (Leverage Gap)**.** *For architectures with equal capabilities, the modification cost ratio equals the inverse leverage ratio:*

$$\frac{ModCost(A_2)}{ModCost(A_1)} = \frac{DOF(A_2)}{DOF(A_1)} = \frac{L(A_1)}{L(A_2)}$$

(L: `L30`)

Theorem 3.12 (Testable Prediction)**.** *If architecture A_1 has $n\times$ lower DOF than A_2 (for equal capabilities), then A_1 requires $n\times$ fewer expected modifications. This is empirically testable against PR/commit data.*

(L: `L42`)

Corollary 3.13 (DOF Ratio Predicts Error Ratio)**.** *The ratio of expected errors between two architectures equals the ratio of their DOF:*

$$\frac{\mathbb{E}[errors(A_2)]}{\mathbb{E}[errors(A_1)]} = \frac{DOF(A_2)}{DOF(A_1)}$$

(L: `L21`)

These theorems transform architectural intuitions into testable hypotheses. A claim that "Pattern X is 3× better than Pattern Y" can be verified by comparing DOF and measuring modification frequency in real codebases.

3.7 Formalization

Formalized in `Leverage/Probability.lean`:

- `dof_reliability_isomorphism`: Theorem 3.9 (the central isomorphism)
- `isomorphism_preserves_failure_ordering`: Ordering preservation
- `isomorphism_roundtrip`: Invertibility proof

- `ordering_equivalence_exact`: Theorem 3.10 (exact ordering equivalence)
- `linear_model_preserves_ordering`: Linear ordering support
- `leverage_gap`: Theorem 3.11
- `testable_modification_prediction`: Theorem 3.12
- `dof_ratio_predicts_error_ratio`: Corollary 3.13
- `lower_dof_lower_errors`: Corollary 3.6
- `expected_errors_from_linearity`: Theorem 3.7
- `ssot_minimal_errors`: SSOT minimality

4 Main Theorems

We prove the core results connecting leverage to error probability and architectural optimality. The theoretical structure is:

1. **DOF-Reliability Isomorphism** (Theorem 3.9): Maps architecture to reliability theory
2. **Leverage-Error Tradeoff** (Theorem 4.2): Connects leverage to error probability
3. **Physical Edit-Energy Floor** (Theorem 4.3): DOF controls minimum edit-energy
4. **Optimality Criterion** (Theorem 4.8): Correctness in a fixed capability class
5. **Composition Stability** (Theorem 4.9): Composition preserves leverage lower bounds

4.1 Recap: DOF-Reliability Isomorphism

The core theoretical contribution (stated in Section 1.3) is that DOF maps formally to series system components. This enables all subsequent results by connecting software architecture to the mature mathematical framework of reliability theory.

Key properties of the isomorphism:

- **Preserves ordering:** If $\text{DOF}(A_1) < \text{DOF}(A_2)$, then $P_{\text{error}}(A_1) < P_{\text{error}}(A_2)$
- **Invertible:** An architecture can be reconstructed from its series system representation
- **Compositional:** $\text{DOF}(A_1 \oplus A_2) = \text{DOF}(A_1) + \text{DOF}(A_2)$ (series systems combine)

4.2 The Leverage Maximization Principle

Given the DOF-Reliability Isomorphism, the following is a *corollary*:

Theorem 4.1 (Leverage Maximization Principle)**.** *For any architectural decision with alternatives $A_1, \ldots, A_n$ meeting capability requirements, the optimal choice maximizes leverage:*

$$A^* = \arg\max_{A_i} L(A_i) = \arg\max_{A_i} \frac{|\mathit{Capabilities}(A_i)|}{\mathit{DOF}(A_i)}$$

(L: `L31`, `L28`)

Note: This is *not* the central theorem; it is a consequence of the DOF-Reliability Isomorphism. The isomorphism is the deep result; "maximize leverage" is the actionable heuristic derived from it.

4.3 Leverage-Error Tradeoff

Theorem 4.2 (Leverage-Error Tradeoff)**.** *For architectures A_1, A_2 with equal capabilities:*

$$Cap(A_1) = Cap(A_2) \wedge L(A_1) > L(A_2) \implies P_{error}(A_1) < P_{error}(A_2)$$

(L: L29)

Proof. Given: $\text{Cap}(A_1) = \text{Cap}(A_2)$ and $L(A_1) > L(A_2)$.

Since $L(A) = |\text{Cap}(A)|/\text{DOF}(A)$ and capabilities are equal:

$$\frac{|\text{Cap}(A_1)|}{\text{DOF}(A_1)} > \frac{|\text{Cap}(A_2)|}{\text{DOF}(A_2)}$$

With $|\text{Cap}(A_1)| = |\text{Cap}(A_2)|$:

$$\frac{1}{\text{DOF}(A_1)} > \frac{1}{\text{DOF}(A_2)} \implies \text{DOF}(A_1) < \text{DOF}(A_2)$$

By Corollary 3.6:

$$\text{DOF}(A_1) < \text{DOF}(A_2) \implies P_{\text{error}}(A_1) < P_{\text{error}}(A_2)$$

■

Corollary: Maximizing leverage minimizes error probability (for fixed capabilities).

4.4 Physical Edit-Energy Floor

Theorem 4.3 (Physical Edit-Energy Floor)**.** *Fix a physical edit model $\mathcal{M}$ with per-edit conversion constant $j_{\mathcal{M}} > 0$. For any architecture A:*

$$E_{\min}(A; \mathcal{M}) = j_{\mathcal{M}} \cdot \text{DOF}(A).$$

Hence $\text{DOF}(A_1) < \text{DOF}(A_2)$ *implies* $E_{\min}(A_1; \mathcal{M}) < E_{\min}(A_2; \mathcal{M})$.

(L: L1 , L5)

Proof. In the mechanized model, modification complexity is definitionally DOF. Multiplying by a positive per-edit conversion constant gives the lower bound and strict monotonicity in DOF. ■

Corollary 4.4 (Leverage-Energy Monotonicity in a Capability Class)**.** *For architectures A_1, A_2 with equal capabilities, if $L(A_1) > L(A_2)$ then*

$$E_{\min}(A_1; \mathcal{M}) < E_{\min}(A_2; \mathcal{M}),$$

and the induced energy gap is strictly positive.

(L: L3 , L6)

Theorem 4.5 (Budget Feasibility Boundary)**.** *For physical model $\mathcal{M}$ and budget B, implementation feasibility is equivalent to clearing the floor:*

$$E_{\min}(A; \mathcal{M}) \leq B \iff \textit{feasible}(A, B, \mathcal{M}).$$

Hence $B < E_{\min}(A; \mathcal{M})$ implies infeasibility. Also, for fixed B and equal capabilities, if $L(A_1) > L(A_2)$ and A_2 is feasible under B, then A_1 is feasible under B.

(L: L2 , L4)

Corollary 4.6 (Physical Assumption Necessity Witnesses)**.** *The physical layer uses two explicit premises for no-go style infeasibility claims: positive per-edit conversion and an external budget bound. Dropping positivity admits a zero-floor countermodel. Dropping the external budget-bound premise blocks infeasibility conclusions because a feasible budget witness always exists.*

(L: L8 , L7)

4.5 Metaprogramming Dominance

Theorem 4.7 (Metaprogramming Dominance)**.** *For any $N \in \mathbb{N}$, there exists a metaprogramming architecture M with DOF$(M) = 1$ and $L(M) \geq N$.*

(L: L36 , L35)

Proof. Let M be a metaprogramming architecture with a single source definition (DOF$(M) = 1$) and at least N derived capabilities. Then $L(M) = |\text{Cap}(M)|/1 \geq N$.

Modeling note. In any fixed finite model, $L(M)$ is a positive rational. The colloquial claim "leverage $= \infty$" means: for any bound N, the architecture can be extended to achieve $L \geq N$ while keeping DOF $= 1$. The Lean formalization proves the for-all-N version (L36). ■

4.6 Architectural Decision Criterion

Theorem 4.8 (Optimal Architecture)**.** *Given requirements R, architecture A^* is optimal in its capability class if and only if:*

1. *Cap$(A^*) \supseteq R$ (feasibility)*
2. *$\forall A'$ with Cap$(A') =$ Cap(A^*) and Cap$(A') \supseteq R$: $L(A^*) \geq L(A')$ (maximality in fixed capability class)*

(L: L38 , L34)

Proof. ($\Leftarrow$) Suppose A^* satisfies (1) and (2). Then A^* is feasible and has maximum leverage among feasible architectures in the same capability class. By Theorem 4.2, this minimizes error probability in that class.

($\Rightarrow$) If (1) fails, A^* is infeasible. If (2) fails, there exists A' in the same capability class with $L(A') > L(A^*)$, so $P_{\text{error}}(A') < P_{\text{error}}(A^*)$ by Theorem 4.2, contradicting optimality. ■

Decision Procedure:

1. Enumerate candidate architectures $\{A_1, \ldots, A_n\}$
2. Filter: Keep only A_i with Cap$(A_i) \supseteq R$
3. Optimize: Choose $A^* = \arg\max_i L(A_i)$

4.7 Leverage Composition

Theorem 4.9 (Leverage Composition)**.** *For modular architecture $A = A_1 \oplus A_2$ with disjoint components:*

1. *$DOF(A) = DOF(A_1) + DOF(A_2)$*
2. *if $L(A_1) \geq 1$ and $L(A_2) \geq 1$, then $L(A) \geq 1$*

(L: `L19`, `L18`)

Proof. (1) By Proposition 2.4.

(2) If $L(A_i) \geq 1$, then $|\mathrm{Cap}(A_i)| \geq \mathrm{DOF}(A_i)$ for $i = 1, 2$. Summing both inequalities gives

$$|\mathrm{Cap}(A_1)| + |\mathrm{Cap}(A_2)| \geq \mathrm{DOF}(A_1) + \mathrm{DOF}(A_2),$$

which is exactly $L(A) \geq 1$ under additive composition. ■

Interpretation: Composition preserves a baseline leverage floor under the stated assumptions.

Remark 4.10 (Composition Breaks SSOT)*.* Theorem 4.9 preserves leverage *floors*, but composing two SSOT architectures breaks SSOT: if $\mathrm{DOF}(A_1) = \mathrm{DOF}(A_2) = 1$, then $\mathrm{DOF}(A_1 \oplus A_2) = 2$. This is formalized as `compose_breaks_ssot` in `Leverage/BridgeToDQ.lean`. The composition tax is unavoidable: distributed systems consisting of k independent SSOT components have $\mathrm{DOF} = k$, placing them in the coNP-hard regime ($\mathrm{srank} = k > 1$) with mandatory thermodynamic cost per decision cycle.

4.8 Formalization

Main optimization theorems are formalized in `Leverage/Theorems.lean`; physical edit-energy theorems are formalized in `Leverage/Physical.lean`:

- `L29`: Theorem 4.2
- `L1` and `L5`: Theorem 4.3
- `L3` and `L6`: Corollary 4.4
- `L2` and `L4`: Theorem 4.5
- `L7` and `L8`: Corollary 4.6
- `L35` and `L36`: Theorem 4.7
- `L34` and `L38`: Theorem 4.8
- `L18` and `L19`: Theorem 4.9
- `Leverage/Physical.lean`: physical edit-energy floor theorems

4.9 Integration with Dependencies

The leverage framework provides the unifying theory for results proved in `AbstractClassSystem` and `Ssot`:

Theorem 4.11 (`AbstractClassSystem` as Leverage Instance)**.** *Nominal typing dominance from* `AbstractClassSystem` *[24] is an instance of leverage maximization:*

- *Nominal typing adds 4 B-dependent capabilities over duck typing*
- *DOF remains fixed in the mechanized typing instance*
- *Therefore nominal typing has strictly higher leverage*

(L: `L14`, `L13`)

Theorem 4.12 (`Ssot` as Leverage Instance)**.** *SSOT dominance from* `Ssot` *[25] is an instance of leverage maximization:*

- *SSOT fixes DOF at 1 for a structural fact*
- *Non-SSOT replication yields DOF $= n$ for the same fact*
- *Therefore leverage improves by factor n*

(L: `L10`, `L11`)

These integration theorems are formalized in:

- `Leverage/Typing.lean`
- `Leverage/SSOT.lean`

5 Five-Way Equivalence

This section establishes the central result: five independent characterizations of the same architectural property (DOF = 1 / SSOT) derived from five distinct first-principles frameworks. All proofs are machine-checked in Lean 4.

5.1 Framework 1: Engineering Optimization (Leverage)

Theorem 5.1 (Maximum Leverage). For any architecture A with $\text{Cap}(A) > 0$, A achieves maximum leverage among architectures with equal capabilities if and only if $\text{DOF}(A) = 1$.

Formally: $L(A) = \max_{A':\text{Cap}(A')=\text{Cap}(A)} L(A') \iff \text{DOF}(A) = 1$.

An architecture has the highest possible capability-to-DOF ratio exactly when it has a single degree of freedom.

Proof.

- (Forward) If $\text{DOF}(A) = 1$, then $L(A) = |\text{Cap}(A)|/1 = |\text{Cap}(A)|$. Any A' with same capabilities has $\text{DOF}(A') \geq 1$, so $L(A') \leq |\text{Cap}(A)| = L(A)$.
- (Backward) If A has maximum leverage but $\text{DOF}(A) > 1$, construct A' with $\text{DOF}(A') = 1$ and same capabilities. Then $L(A') = |\text{Cap}(A)| > |\text{Cap}(A)|/\text{DOF}(A) = L(A)$, contradiction.

(L: `L31`, `L28`, `L48`)

5.2 Framework 2: Architectural Epistemic Coherence (Ssot)

Theorem 5.2 (Coherence). An architecture satisfies the Single Source of Truth property if and only if $\mathrm{DOF}(A) = 1$.

Formally: $\mathrm{SSOT}(A) \iff \mathrm{DOF}(A) = 1$.

Epistemic coherence (one source of truth per structural fact) is equivalent to having a single degree of freedom.

(L: Proved in `Ssot` *(*`ORA1`*))*

This is the $\mathrm{DOF} = 1$ characterization from the Ssot coherence theorem.

5.3 Framework 3: Information-Theoretic Structural Rank (DecisionQuotient)

Theorem 5.3 (DOF-Structural Rank Isomorphism). For any architecture A, let $\mathrm{canonicalDP}(n)$ be the canonical decision problem with n boolean coordinates. Then:

$$\mathrm{srank}(\mathrm{canonicalDP}(\mathrm{DOF}(A))) = \mathrm{DOF}(A)$$

The structural rank (number of relevant decision coordinates) equals the degrees of freedom. The canonical encoding uses: states as n boolean coordinates, actions as either querying a coordinate or falling back, and utilities that reward correct coordinate identification.

The canonical encoding is:

- States: $\mathrm{Fin}\ n \to \mathrm{Bool}$ (n binary coordinates)
- Actions: $\mathrm{Fin}\ n \oplus \mathrm{Unit}$ (query coordinate i, or fallback)
- Utility: $u(\mathrm{inl}\ i, s) = 2$ if $s(i) = \mathrm{true}$, else 0; $u(\mathrm{inr}\ (), s) = 1$

(L: `L43`*,* `L51`*,* `L46`*)*

Corollary 5.4. $\mathrm{DOF} = 1$ if and only if $\mathrm{srank} = 1$.

Formally: $\mathrm{DOF}(A) = 1 \iff \mathrm{srank}(\mathrm{canonicalDP}(\mathrm{DOF}(A))) = 1$.

An architecture has a single degree of freedom exactly when its canonical decision problem has minimal structural rank.

(L: Immediate from Theorem 5.3)

5.4 Framework 4: Computational Complexity (Tractability)

Theorem 5.5 (Tractability Boundary). For the canonical decision problem with n coordinates:

1. If $n = 1$ ($\mathrm{srank} = 1$), sufficiency checking is in P (polynomial time).
2. If $n > 1$ ($\mathrm{srank} > 1$), sufficiency checking is coNP-hard.

When the structural rank is 1, determining whether current information suffices for an optimal decision is computationally tractable. When the structural rank exceeds 1, the same problem is computationally intractable (coNP-hard).

(L: Proved in `DecisionQuotient` *(L56, L53), imported via* `Leverage/BridgeToDQ.lean`*)*

This connects the architectural property (DOF $= 1$) to computational feasibility of decision-making.

Corollary 5.6. DOF $= 1$ (SSOT) is the unique architecture class for which sufficiency checking is tractable.

Among architectures, only those with a single degree of freedom admit tractable sufficiency checking.

(L: Combine Corollary 5.4 with Theorem 5.5; L47)

5.5 Framework 5: Thermodynamic Selection (Statistical Physics)

Theorem 5.7 (Thermodynamic Selection). Let M be a thermodynamic model with Landauer calibration. For any architecture A with DOF > 1:

1. There exist decision instances where sufficiency checking requires $\Omega(2^{\mathrm{DOF}})$ logical operations.
2. Under the physical axioms of [23] (LP38): Landauer (EP1), nontrivial state space (EP4), finite signal speed (LP43), finite universe energy (LP44) — P $\neq$ NP is proved, hence P $\neq$ coNP; no polynomial-time sufficiency certification exists.
3. **(Unconditional)** Each sufficiency-check cycle incurs mandatory positive energy cost $\geq j_{\mathcal{M}} \cdot \mathrm{srank}(A) > j_{\mathcal{M}}$, proved from Landauer calibration and the DOF $=$ srank identity alone, with no complexity hypothesis.

For DOF $= 1$, the energy lower bound is $j_{\mathcal{M}}$ (the Landauer minimum).

Architectures with more than one degree of freedom necessarily incur thermodynamic costs per decision cycle. Point 3 holds unconditionally: the energy bound follows from Landauer and the DOF = srank identity (L43). Point 2 holds under the physical axioms of [23] (LP38), which prove P $\neq$ NP, hence P $\neq$ coNP.

Physical assumptions (point 3 only):

- Positive Landauer constant ($j_{\mathcal{M}} > 0$) — cf. Theorem 4.4 (L3, L6)
- Landauer calibration ($k_B T \ln 2$ joules per bit)

(L: L54, L55, L49)

Physical edit-energy floor from Section 4.3 is a special case: DOF controls minimum edit-energy, and higher leverage implies lower energy in a fixed capability class (Theorem 4.3, L1, L5).

5.6 The Five-Way Equivalence

Theorem 5.8 (Five-Way Equivalence). For any architecture A with $\mathrm{Cap}(A) > 0$, the following are equivalent:

$$\mathrm{DOF}(A) = 1 \iff \text{max leverage} \iff \mathrm{SSOT}(A) \iff \mathrm{srank}(A) = 1 \iff \text{tractable sufficiency} \iff \text{minimum therm}$$

Five independent scientific frameworks—engineering optimization, epistemic coherence, information geometry, computational complexity, and statistical physics—all select the same architectural property: having exactly one degree of freedom.

Domain	Characterization of DOF $= 1$
Engineering Optimization	Maximum leverage
Architectural Epistemic	Single Source of Truth
Information Geometry	Structural rank $= 1$
Computational Complexity	Tractable sufficiency checking
Statistical Physics	Minimum thermodynamic cost per cycle

Proof. The equivalence follows from Theorems 5.1-5.7:

1. DOF $= 1 \iff$ max leverage: Theorem 5.1 (L31, L28)
2. DOF $= 1 \iff$ SSOT: Theorem 5.2 [25]
3. DOF $=$ srank: Theorem 5.3, so DOF $= 1 \iff$ srank $= 1$
4. srank $= 1 \iff$ tractable sufficiency: Theorem 5.5 [23]. Forward direction (srank $= 1 \to$ polynomial) is unconditional. Backward direction (tractable $\to$ srank $= 1$) holds under P $\neq$ coNP, which follows from the physical axioms of [23] (LP38).
5. DOF $> 1 \iff$ above-minimum thermodynamic cost: Theorem 5.7, so DOF $= 1 \iff$ minimum thermodynamic cost

The transitivity of logical equivalence gives the five-way equivalence.
(L: L44, ORA1, L43, L47, L55)

Corollary 5.9 (Convergence). Within the `DecisionQuotient` canonical decision problem framework, the boolean-coordinate encoding (states as n boolean coordinates, actions as coordinate queries or fallback, utilities rewarding correct identification) is the unique decision structure with $n = 1$ that simultaneously satisfies all five characterizations.

In the canonical decision problem representation, there is exactly one $n = 1$ structure, and all five characterizations select it. This is a consequence of the equivalences, not an additional theorem. Note: there are many architectures with DOF $= 1$ (monolith, single DB table, single endpoint); the uniqueness claim is scoped to the canonical decision problem encoding.

Proof. Each characterization is equivalent to DOF $= 1$ (Theorems 5.1–5.7). Within the canonical encoding, DOF $= 1$ fixes $n = 1$, and Fin $1 \to$ Bool has a unique element. All five requirements therefore select the same structure.
(L: All five requirements force $n = 1$ by the above theorems; uniqueness within the canonical encoding follows from $|\text{Fin } 1 \to \text{Bool}| = 2$)

5.7 Formalization

The following Lean 4 proofs establish the connections:

- `Leverage/Foundations.lean`:
  - L31, L28 (L50): DOF $= 1 \to$ max leverage
  - L48: max leverage $\to$ DOF $= 1$
  - L44: biconditional
- `Leverage/BridgeToDQ.lean`:

  - L43: DOF = srank
  - L51: DOF = 1 → srank = 1
  - L46: DOF > 1 → srank > 1
  - L54: DOF > 1 → mandatory energy
  - L47: max leverage → tractable

- `DecisionQuotient/Tractability/StructuralRank.lean`: srank = 1 → P (L56), srank > 1 → coNP-hard (L53)
- `DecisionQuotient/ThermodynamicLift.lean`: energy lower bounds under Landauer calibration (L49)

The cross-module dependency chain is live: `Leverage` → `DecisionQuotient` → Mathlib.

5.8 Relation to Prior Sections

This section subsumes and extends Section 4's results:

- Theorem 4.1 (Leverage-Error Tradeoff, L29) is a consequence of the equivalence between leverage and DOF
- Theorem 4.2 (Modification Complexity Gap, L30) follows from DOF = 1 minimizing modification cost
- Theorem 4.3 (Optimal Architecture, L38, L34) is strengthened: the optimal architecture is now characterized by five independent properties
- Theorem 4.4 (Metaprogramming Dominance, L36, L35) is a special case: unbounded derivations from a single source achieve DOF = 1

5.9 England Replication Inequality (Proved)

All previously open conjectures are now proved. The England Replication Inequality is mechanized in `Leverage/BridgeToDQ.lean` (L45).

Theorem 5.1 (England Replication Inequality)**.** *Let $\Delta S_{\min}(srank) = srank \cdot k_B \ln 2$ be the minimal entropy production under Landauer calibration. For a single-source architecture ($srank = 1$) and a k-copy replication architecture ($srank = k$):*

$$\Delta S_{\min}(1) + k_B \ln k \leq \Delta S_{\min}(k)$$

equivalently, $\Delta S_{\min}(k) - \Delta S_{\min}(1) \geq k_B \ln k$.

(L: L45)

Proof. The gap is $(k-1) \cdot k_B \ln 2$. Since $k \leq 2^{k-1}$ (L52), taking logs gives $\ln k \leq (k-1)\ln 2$, so the gap is $\geq k_B \ln k$. ■

Modeling note. $\Delta S_{\min}$ is a definition within the model: the exact Landauer entropy cost per cycle, not a lower bound on arbitrary implementations. The "min" refers to physical optimality outside the formalism.

6 Instances

We demonstrate that the leverage framework unifies prior results and applies to diverse architectural decisions.

6.1 Instance 1: Single Source of Truth (SSOT)

We previously formalized the DRY principle, proving that Python uniquely provides SSOT for structural facts via definition-time hooks and introspection. Here we show SSOT is an instance of leverage maximization.

6.1.1 Prior Result

Published Theorem: A language enables SSOT for structural facts if and only if it provides (1) definition-time hooks AND (2) introspectable derivation results. Python is the only mainstream language satisfying both requirements.

Modification Complexity: For structural fact F with n use sites:

- SSOT: $M(\text{change } F) = 1$ (modify source, derivations update automatically)
- Non-SSOT: $M(\text{change } F) = n$ (modify each use site independently)

6.1.2 Leverage Perspective

Definition 6.1 (SSOT Architecture)**.** Architecture A_{SSOT} for structural fact F has:

- Single source S defining F
- Derived use sites updated automatically from S
- DOF $= 1$ (only S is independently modifiable)

Definition 6.2 (Non-SSOT Architecture)**.** Architecture $A_{\text{non-SSOT}}$ for structural fact F with n use sites has:

- n independent definitions (copied or manually synchronized)
- DOF $= n$ (each definition independently modifiable)

Theorem 6.3 (SSOT Leverage Dominance)**.** *For structural fact with n use sites:*

$$\frac{L(A_{SSOT})}{L(A_{non\text{-}SSOT})} = n$$

(L: `L11`, `L9`)

Proof. Both architectures provide same capabilities: $|\text{Cap}(A_{\text{SSOT}})| = |\text{Cap}(A_{\text{non-SSOT}})| = c$.

DOF:

$$\text{DOF}(A_{\text{SSOT}}) = 1$$
$$\text{DOF}(A_{\text{non-SSOT}}) = n$$

Leverage:

$$L(A_{\text{SSOT}}) = c/1 = c$$
$$L(A_{\text{non-SSOT}}) = c/n$$

Ratio:

$$\frac{L(A_{\text{SSOT}})}{L(A_{\text{non-SSOT}})} = \frac{c}{c/n} = n$$

■

Corollary 6.4 (Unbounded Advantage)**.** *As use sites grow ($n \to \infty$), leverage advantage grows unbounded.*

Corollary 6.5 (Error Probability)**.** *For small p:*

$$\frac{P_{error}(A_{non\text{-}SSOT})}{P_{error}(A_{SSOT})} \approx n$$

Connection to Prior Work: Our published Theorem 6.3 (Unbounded Complexity Gap) showed $M(\text{SSOT}) = O(1)$ vs $M(\text{non-SSOT}) = \Omega(n)$. Theorem 6.3 provides the leverage perspective: SSOT achieves n-times better leverage.

6.2 Instance 2: Nominal Typing Dominance

We previously proved nominal typing strictly dominates structural and duck typing for OO systems with inheritance. Here we show this is an instance of leverage maximization.

6.2.1 Prior Result

Published Theorems:

1. Theorem 3.13 (Provenance Impossibility): No shape discipline can compute provenance
2. Theorem 3.19 (Capability Gap): Gap = B-dependent queries = {provenance, identity, enumeration, conflict resolution}
3. Theorem 3.5 (Strict Dominance): Nominal strictly dominates duck typing

6.2.2 Leverage Perspective

Definition 6.6 (Typing Discipline as Architecture)**.** A typing discipline D is an architecture where:

- Components = type checker, runtime dispatch, introspection APIs
- Capabilities = queries answerable by the discipline

Duck Typing: Uses only Shape axis (S: methods, attributes)

- Capabilities: Shape checking ("Does object have method m?")
- Cannot answer: provenance, identity, enumeration, conflict resolution

Nominal Typing: Uses Name + Bases + Shape axes ($N + B + S$)

- Capabilities: All duck capabilities PLUS 4 B-dependent capabilities
- Can answer: "Which type provided method m?" (provenance), "Is this exactly type T?" (identity), "List all subtypes of T" (enumeration), "Which method wins in diamond?" (conflict)

Theorem 6.7 (Nominal Leverage Dominance)**.** *Assuming* DOF(Nominal) = DOF(Duck) = d *(equal implementation cost hypothesis):*

$$L(Nominal) > L(Duck)$$

(L: L13, L12)

Proof. Let $c_{\text{duck}} = |\text{Cap(Duck)}|$ and $c_{\text{nominal}} = |\text{Cap(Nominal)}|$.

By Theorem 3.19 (published): $c_{\text{nominal}} = c_{\text{duck}} + 4$.

By hypothesis: DOF(Nominal) = DOF(Duck) = d.

Therefore:

$$L(\text{Nominal}) = \frac{c_{\text{duck}} + 4}{d} > \frac{c_{\text{duck}}}{d} = L(\text{Duck})$$

Status of hypothesis. The equal-DOF assumption is formalized as an explicit premise in the Lean mechanization (L13). Whether it holds in practice depends on the implementation. Abstractly: the typing discipline is one component; DOF is the number of independent change points in that component. The hypothesis says adding name-based dispatch (N + B axes) does not introduce more independent change points than shape-based dispatch (S axis alone). ∎

Connection to Prior Work: Our published Theorem 3.5 (Strict Dominance) showed nominal typing provides strictly more capabilities for same DOF cost. Theorem 6.7 provides the leverage formulation.

6.3 Instance 3: Microservices Architecture

Should a system use microservices or a monolith? How many services are optimal? The leverage framework provides answers. This architectural style, popularized by Fowler and Lewis [10], traces its roots to the Unix philosophy of "doing one thing well" [21].

6.3.1 Architecture Comparison

Monolith:

- Components: Single deployment unit
- DOF = 1
- Capabilities: Basic functionality, simple deployment

n **Microservices:**

- Components: n independent services
- DOF = n (each service independently deployable/modifiable)
- Additional Capabilities: Independent scaling, independent deployment, fault isolation, team autonomy, polyglot persistence [10]

6.3.2 Leverage Analysis

Let c_0 = capabilities provided by monolith.

Let $\Delta c = 5$ denote the additional capabilities from microservices: independent scaling, independent deployment, fault isolation, team autonomy, and polyglot persistence.

Leverage:

$$L(\text{Monolith}) = c_0/1 = c_0$$
$$L(n \text{ Microservices}) = (c_0 + \Delta c)/n = (c_0 + 5)/n$$

Break-even Point:

$$L(\text{Microservices}) \geq L(\text{Monolith}) \iff \frac{c_0 + 5}{n} \geq c_0 \iff n \leq 1 + \frac{5}{c_0}$$

Interpretation: If base capabilities $c_0 = 5$, then $n \leq 2$ services is optimal. For $c_0 = 20$, up to $n = 1.25$ (i.e., monolith still better). Microservices justified only when additional capabilities significantly outweigh DOF cost.

6.4 Instance 4: REST API Design

Generic endpoints vs specific endpoints: a leverage tradeoff.

6.4.1 Architecture Comparison

Specific Endpoints: One endpoint per use case

- Example: `GET /users`, `GET /posts`, `GET /comments`, ...
- For n use cases: DOF $= n$
- Capabilities: Serve n use cases

Generic Endpoint: Single parameterized endpoint

- Example: `GET /resources/:type/:id`
- DOF $= 1$
- Capabilities: Serve n use cases (same as specific)

6.4.2 Leverage Analysis

$$L(\text{Generic}) = n/1 = n$$
$$L(\text{Specific}) = n/n = 1$$

Advantage: $L(\text{Generic})/L(\text{Specific}) = n$

Tradeoff: Generic endpoint has higher leverage but may sacrifice:

- Type safety (dynamic routing)
- Specific validation per resource
- Tailored response formats

Decision Rule: Use generic if $n > k$ where k is complexity threshold (typically $k \approx$ 3–5).

6.5 Instance 5: Configuration Systems

Convention over configuration (CoC) reduces developer decisions and preserves flexibility [14]. In this framework it is leverage maximization via defaults.

6.5.1 Architecture Comparison

Explicit Configuration: Must set all m parameters

- DOF $= m$ (each parameter independently set)
- Capabilities: Configure m aspects

Convention over Configuration: Provide defaults, override only k parameters

- DOF $= k$ where $k \ll m$
- Capabilities: Configure same m aspects (defaults handle rest)

Example (Rails vs Java EE):

- Rails: 5 config parameters (convention for rest)
- Java EE: 50 config parameters (explicit for all)

6.5.2 Leverage Analysis

$$L(\text{Convention}) = m/k$$
$$L(\text{Explicit}) = m/m = 1$$

Advantage: $L(\text{Convention})/L(\text{Explicit}) = m/k$
For Rails example: $m/k = 50/5 = 10$ ($10\times$ leverage improvement).

6.6 Instance 6: Database Schema Normalization

Normalization eliminates redundancy, maximizing leverage. This concept, introduced by Codd [7], is the foundation of relational database design [9].

6.6.1 Architecture Comparison

Consider customer address stored in database:
Denormalized (Address in 3 tables):

- `Users` table: address columns
- `Orders` table: shipping address columns
- `Invoices` table: billing address columns
- DOF $= 3$ (address stored 3 times)

Normalized (Address in 1 table):

- `Addresses` table: single source
- Foreign keys from `Users`, `Orders`, `Invoices`
- DOF $= 1$ (address stored once)

6.6.2 Leverage Analysis

Both provide same capability: store/retrieve addresses.

$$L(\text{Normalized}) = c/1 = c$$
$$L(\text{Denormalized}) = c/3$$

Advantage: $L(\text{Normalized})/L(\text{Denormalized}) = 3$
Modification Complexity:

- Change address format: Normalized $M = 1$, Denormalized $M = 3$
- Error probability: $P_{\text{denorm}} = 3p$ vs $P_{\text{norm}} = p$

Tradeoff: Normalization increases leverage but may sacrifice query performance (joins required).

6.7 Summary of Instances

Instance	High Leverage	Low Leverage	Ratio
SSOT	DOF $= 1$	DOF $= n$	n
Nominal Typing	$c + 4$ caps, DOF d	c caps, DOF d	$(c + 4)/c$
Microservices	Monolith (DOF $= 1$)	n services (DOF $= n$)	$n/(c_0 + 5)$
REST API	Generic (DOF $= 1$)	Specific (DOF $= n$)	n
Configuration	Convention (DOF $= k$)	Explicit (DOF $= m$)	m/k
Database	Normalized (DOF $= 1$)	Denormalized (DOF $= n$)	n

Pattern: High leverage architectures achieve n-fold improvement where n is the consolidation factor (use sites, services, endpoints, parameters, or redundant storage).

7 Practical Demonstration

We demonstrate the leverage framework by showing how DOF collapse patterns manifest in OpenHCS, a production 45K LoC Python bioimage analysis platform. This section presents qualitative before/after examples illustrating the leverage archetypes, with PR #44 providing a publicly verifiable anchor.

7.1 The Leverage Mechanism

For a before/after pair $A_{\text{pre}}, A_{\text{post}}$, the **structural leverage factor** is:

$$\rho := \frac{\text{DOF}(A_{\text{pre}})}{\text{DOF}(A_{\text{post}})}.$$

If capabilities are preserved, leverage scales exactly by ρ. The key insight: when $\text{DOF}(A_{\text{post}}) = 1$, we achieve $\rho = n$ where n is the original DOF count.

What counts as a DOF? Independent *definition loci*: manual registration sites, independent override parameters, separately defined endpoints/handlers/rules, duplicated schema/format definitions. The unit is "how many places can drift apart," not lines of code.

7.2 Verifiable Example: PR #44 (Contract Enforcement)

PR #44 ("UI Anti-Duck-Typing Refactor") in the OpenHCS repository provides a publicly verifiable demonstration of DOF collapse:

Before (duck typing): The `ParameterFormManager` class used scattered `hasattr()` checks throughout the codebase. Each dispatch point was an independent DOF: a location that could drift, contain typos, or miss updates when widget interfaces changed.

After (nominal ABC): A single `AbstractFormWidget` ABC defines the contract. All dispatch points collapsed to one definition site. The ABC provides fail-loud validation at class definition time rather than fail-silent behavior at runtime.

Leverage interpretation: DOF collapsed from n scattered dispatch points to 1 centralized ABC. By Theorem 3.1, this achieves $\rho = n$ leverage improvement. The specific value of n is verifiable by inspecting the PR diff.

7.3 Leverage Archetypes

The framework identifies recurring patterns where DOF collapse occurs:

7.3.1 Archetype 1: SSOT (Single Source of Truth)

Pattern: Scattered definitions → single authoritative source.

Mechanism: Metaclass auto-registration, decorator-based derivation, or introspection-driven generation eliminates manual synchronization.

Before: Define class + register in dispatch table (2 loci per type). **After:** Define class; metaclass auto-registers (1 locus per type).

Leverage: $\rho = 2$ per type; compounds across n types.

7.3.2 Archetype 2: Convention over Configuration

Pattern: Explicit parameters → sensible defaults with override.

Mechanism: Framework provides defaults; users override only non-standard values.

Before: Specify all m configuration parameters explicitly. **After:** Override only $k \ll m$ parameters; defaults handle rest.

Leverage: $\rho = m/k$.

7.3.3 Archetype 3: Generic Abstraction

Pattern: Specific implementations → parameterized generic.

Mechanism: Factor common structure into generic endpoint/handler with parameters for variation.

Before: n specific endpoints with duplicated logic. **After:** 1 generic endpoint with n parameter instantiations.

Leverage: $\rho = n$.

7.3.4 Archetype 4: Centralization

Pattern: Scattered cross-cutting concerns → centralized handler.

Mechanism: Middleware, decorators, or aspect-oriented patterns consolidate error handling, logging, authentication, etc.

Before: Each call site handles concern independently. **After:** Central handler; call sites delegate.

Leverage: $\rho = n$ where n is number of call sites.

7.4 Summary

The leverage framework identifies a common mechanism across diverse refactoring patterns: DOF collapse yields proportional leverage improvement. Whether the pattern is SSOT, convention-over-configuration, generic abstraction, or centralization, the mathematical structure is identical: reduce DOF while preserving capabilities.

PR #44 provides a verifiable anchor demonstrating this mechanism in practice. The qualitative value lies not in aggregate statistics but in the *mechanism*: once understood, the pattern applies wherever scattered definitions can be consolidated.

8 Related Work

8.1 Software Architecture Metrics

Coupling and Cohesion [26]: Introduced coupling (inter-module dependencies) and cohesion (intra-module relatedness). Recommend high cohesion, low coupling.

Difference: Our framework is capability-aware. High cohesion correlates with high leverage (focused capabilities per module), but we formalize the connection to error probability.

Cyclomatic Complexity [19]: Counts decision points in code. Higher values are commonly used as a risk indicator, though empirical studies on defect correlation show mixed results.

Difference: Complexity measures local control flow; leverage measures global architectural DOF. Orthogonal concerns.

8.2 Design Patterns

Gang of Four [11]: Catalogued 23 design patterns (Singleton, Factory, Observer, etc.). Patterns codify best practices but lack formal justification.

Connection: Many patterns maximize leverage:

- **Factory Pattern:** Centralizes object creation (DOF = 1 for creation logic)
- **Strategy Pattern:** Encapsulates algorithms (DOF = 1 per strategy family)
- **Template Method:** Defines algorithm skeleton (DOF = 1 for structure)

Our framework explains *why* these patterns work: they maximize leverage.

8.3 Technical Debt

Cunningham [8]: Introduced technical debt metaphor. Poor design creates "debt" that must be "repaid" later.

Connection: Low leverage = high technical debt. Scattered DOF (non-SSOT, denormalized schemas, specific endpoints) create debt. High leverage architectures minimize debt.

8.4 Formal Methods in Software Architecture

Architecture Description Languages (ADLs): Wright [1], ACME [13], Aesop [12]. Formalize architecture structure but not decision-making. See also Shaw and Garlan [22].

Difference: ADLs describe architectures; our framework prescribes optimal architectures via leverage maximization.

ATAM and CBAM: Architecture Tradeoff Analysis Method [15] and Cost Benefit Analysis Method [2]. Evaluate architectures against quality attributes (performance, modifiability, security). See also Bass et al. [4].

Difference: ATAM is qualitative; our framework provides quantitative optimization criterion (maximize L).

Necessity Specifications: Mackay et al. [17] formalize *necessity specifications*: robustness guarantees that modules do only what their specification requires, even under adversarial clients. Soundness is mechanized in Coq.

Connection: Necessity specifications address *behavioral minimality*: modules commit to no more behavior than required. Our framework addresses *structural minimality*: architectures commit to no more DOF than required. Both derive minimal commitments from requirements and prove sufficiency.

8.5 Software Metrics Research

Chidamber-Kemerer Metrics [6]: Object-oriented metrics (WMC, DIT, NOC, CBO, RFC, LCOM). Empirical validation studies [3] found these metrics correlate with external quality attributes.

Connection: Metrics like CBO (Coupling Between Objects) and LCOM (Lack of Cohesion) correlate with DOF. High CBO $\implies$ high DOF. Our framework provides theoretical foundation.

8.6 Metaprogramming and Reflection

Reflection [18]: Languages with reflection enable introspection and intercession. Essential for metaprogramming.

Connection: Reflection enables high leverage (SSOT). Our prior work showed Python's definition-time hooks + introspection uniquely enable SSOT for structural facts.

Metaclasses [5, 16]: Early metaobject and metaclass machinery formalized in CommonLoops; the Metaobject Protocol codified in Kiczales et al.'s AMOP. Enable metaprogramming patterns.

Application: Metaclasses are high-leverage mechanism (DOF = 1 for class structure, unlimited derivations).

9 Extension: Weighted Leverage

The basic leverage framework treats all errors equally. In practice, different decisions carry different consequences. This section extends our framework with *weighted leverage* to capture heterogeneous error severity.

9.1 Weighted Decision Framework

Definition 9.1 (Weighted Decision)**.** A **weighted decision** extends an architecture with:

- **Importance weight** $w \in \mathbb{N}^+$: the relative severity of errors in this decision

- **Risk-adjusted DOF**: $\text{DOF}_w = \text{DOF} \times w$

The key insight is that a decision with importance weight w carries w times the error consequence of a unit-weight decision. This leads to:

Definition 9.2 (Weighted Leverage).

$$L_w = \frac{\text{Capabilities} \times w}{\text{DOF}_w} = \frac{\text{Capabilities}}{\text{DOF}}$$

The cancellation is intentional: weighted leverage preserves comparison properties while enabling risk-adjusted optimization.

9.2 Key Theorems

Theorem 9.3 (Weighted Pareto Optimality). *For any weighted decision d with DOF = 1: d is Pareto-optimal (not dominated by any alternative with higher weighted leverage).*

Proof. Suppose d has DOF = 1. For any d' to dominate d, we would need d'.DOF < 1. But DOF ≥ 1 by definition, so no such d' exists. ■

Theorem 9.4 (Weighted Leverage Transitivity). *$\forall a, b, c$: if a has higher weighted leverage than b, and b has higher weighted leverage than c, then a has higher weighted leverage than c.*

Proof. By algebraic manipulation of cross-multiplication inequalities. Formally verified in Lean (38-line proof). ■

9.3 Practical Application: Feature Flags

Consider two approaches to feature toggle implementation:

Low Leverage (Scattered Conditionals):

- DOF: One per feature × one per use site ($n \times m$)
- Risk: Inconsistent behavior if any site is missed
- Weight: High (user-facing inconsistency)

High Leverage (Centralized Configuration):

- DOF: One per feature
- Risk: Single source of truth eliminates inconsistency
- Weight: Same importance, but $m\times$ fewer DOF

Weighted leverage ratio: $L_{\text{centralized}}/L_{\text{scattered}} = m$, the number of use sites.

9.4 Connection to Main Theorems

The weighted framework preserves all results from Sections 3–5:

- **Theorem 3.1 (Leverage-Error Tradeoff)**: Holds with weighted errors
- **Theorem 3.2 (Metaprogramming Dominance)**: Weight amplifies the advantage
- **Theorem 3.4 (Optimality)**: Weighted optimization finds risk-adjusted optima
- **SSOT Dominance**: Weight w makes $n \times w$ leverage advantage

All proofs verified in Lean: `Leverage/WeightedLeverage.lean` (348 lines, 0 sorry placeholders).

10 Conclusion

10.1 Methodology and Disclosure

Role of LLMs in this work. This paper was developed through human-AI collaboration. The author provided the core insight (that DOF = 1 is selected by five independent scientific frameworks) while large language models (Claude, GPT-4) served as implementation partners for formalization, proof drafting, and LaTeX generation.

The Lean 4 proofs (48509 lines, 0 `sorry` placeholders) were iteratively developed: the author specified theorems, the LLM proposed proof strategies, and the Lean compiler verified correctness. Machine-checked proofs are correct regardless of generation method.

What the author contributed: The five-way convergence insight, the identification of structural rank as the information-geometric coordinate of DOF, the thermodynamic selection theorem, the cross-paper dependency chain, the open conjectures, and the OpenHCS case study selection.

What LLMs contributed: LaTeX drafting, Lean tactic suggestions, prose refinement, and exploration of proof strategies.

10.2 Summary

The central result is the Five-Way Equivalence (Theorem 5.8): five independent scientific frameworks all characterize the single-source condition (DOF = 1).

Framework	**DOF = 1 means**	**Source**
Engineering optimization	Maximum leverage $L = \lvert\text{Cap}\rvert/\text{DOF}$	`Leverage`
Epistemic coherence	Single Source of Truth	`Ssot`
Information geometry	Structural rank = 1	`DecisionQuotient`
Computational complexity	Tractable sufficiency checking	`DecisionQuotient`
Statistical physics	Minimum thermodynamic cost per cycle	`Leverage`, `DecisionQuotient`

The engineering consequences (Theorems 1 to 6 from the prior conclusion) are corollaries: DOF-Reliability Isomorphism, Leverage-Error Tradeoff, Modification Complexity Gap, Physical Edit-Energy Floor, Budget Feasibility Boundary, and the Optimal Architecture decision procedure. All are machine-checked in Lean 4 with live cross-module imports (`Leverage` → `DecisionQuotient` → `Ssot` → Mathlib).

What is new in `Leverage` relative to `AbstractClassSystem` and `Ssot`:

- The convergence theorem itself: `AbstractClassSystem` and `Ssot` contain two of the five equivalences; this paper closes the chain.
- The thermodynamic selection theorem (L55 in `BridgeToDQ.lean`): mandatory energy under Landauer calibration for DOF > 1, proved unconditionally.
- The identification of structural rank as the information coordinate for DOF.
- The England Replication Inequality (L45): $\Delta S_{\min}(k) - \Delta S_{\min}(1) \geq k_B \ln k$, giving the thermodynamic advantage of SSOT a quantitative non-equilibrium interpretation. No open conjectures remain.

10.3 No Open Conjectures

All conjectures from the original submission are now proved in `Leverage/BridgeToDQ.lean`:

- `L55`: unconditional energy bound, no P $\neq$ coNP hypothesis.
- `L49`: quantitative Landauer-linear energy bound.
- `L45`: $\Delta S_{\min}(k) - \Delta S_{\min}(1) \geq k_B \ln k$ (Theorem 5.1).

10.4 Decision Procedure

For practitioners, the five-way equivalence implies a principled architectural decision procedure.

Given requirements R, choose optimal architecture via:

1. **Enumerate:** List candidate architectures $\{A_1, \ldots, A_n\}$
2. **Filter:** Keep only A_i with $\mathrm{Cap}(A_i) \supseteq R$
3. **Compute:** Calculate $L(A_i) = |\mathrm{Cap}(A_i)|/\mathrm{DOF}(A_i)$ for each
4. **Optimize:** Choose $A^* = \arg\max_i L(A_i)$

Justification: By the Five-Way Equivalence, A^* simultaneously (a) maximizes leverage, (b) satisfies SSOT, (c) has minimum structural rank, (d) admits tractable sufficiency checking, and (e) minimizes thermodynamic cost. These are five independent validations of the same choice.

10.5 Limitations

1. Independence Assumption: The probability model treats DOF-level errors as independent under the orthogonality assumptions from `AbstractClassSystem` [24]. Real systems may have correlated failure modes not captured by the isomorphism.

2. Constant Error Rate: Assumes p is uniform across components. Some components are more error-prone than others; a weighted version is future work.

3. P $\neq$ coNP Hypothesis (resolved): The thermodynamic energy bound in Theorem 5.7, point 3 is now proved unconditionally (`L55`). Point 2 (complexity hardness) holds under P $\neq$ coNP, which follows from the physical axioms of [23] (`LP38`).

4. Capability Quantification: We count capabilities qualitatively. Some capabilities are more valuable; a weighted leverage extension $L = \sum w_i c_i/\mathrm{DOF}$ is natural but unformalized.

10.6 Impact

For Physicists: The England Replication Inequality (Theorem 5.1) is proved from counting: $|\mathrm{Fin}\ k \to \mathrm{Bool}| = 2^k$, combined with $k \leq 2^{k-1}$ (induction), yields $\Delta S_{\min}(k) - \Delta S_{\min}(1) \geq k_B \ln k$. The only physics input is Landauer's constant k_B. This provides a mechanized derivation of a non-equilibrium thermodynamic bound. Additionally, P $\neq$ NP is proved from physical law (`LP38`: Landauer, finite energy, finite signal speed, nontrivial state space) in 1250+ lines of Lean 4.

For Information Theorists: The structural rank (srank) equals the DOF, and srank = 1 is equivalent to tractable sufficiency checking. The Five-Way Equivalence connects information geometry to computational complexity and thermodynamics.

For Software Practitioners: Five independent reasons to prefer DOF = 1 architectures. When choosing between alternatives, compute leverage and select maximum. Doing so simultaneously satisfies epistemic coherence, minimizes structural rank, enables tractable decision-making, and minimizes thermodynamic cost.

10.7 Final Remarks

This paper proves thermodynamic bounds on information-processing systems from first principles. The single-source condition (DOF = 1) is characterized as optimal by five independent frameworks: engineering, epistemics, information theory, computational complexity, and statistical physics.

All theorems are machine-checked. The thermodynamic bound is unconditional; the England Replication Inequality gives the SSOT entropy advantage as $k_B \ln k$ per update cycle. The only explicit physics axioms are the Second Law (non-negative entropy production) and the Thermodynamic Uncertainty Relation (Barato-Seifert 2015).

A Lean Proof Artifacts

This appendix reports machine-check status and proof traceability directly from source and generated mapping artifacts.

A.1 Verification Status

Lean summary: 48509 lines, 2079 theorems/lemmas, 0 `sorry`, across 195 files.

File	Lines	Theorems/Lemmas
`LambdaDR.lean`	343	24
`Leverage.lean`	115	0
`Leverage/Foundations.lean`	234	13
`Leverage/Probability.lean`	841	39
`Leverage/Theorems.lean`	303	20
`Leverage/SSOT.lean`	192	13
`Leverage/Typing.lean`	210	15
`Leverage/Examples.lean`	184	4
`Leverage/WeightedLeverage.lean`	348	13
`Leverage/BridgeToDQ.lean`	433	26
`Leverage/FiveWayEquivalence.lean`	149	7
`Leverage/CrossPaperDependencies.lean`	332	22
`lakefile.lean`	19	0
Total	**48509**	**2079**

Build command: `cd proofs && lake build`

A.2 Claim Coverage Matrix

Paper handle	Status	Lean support
`cor:dof-errors`	Full	`L.architecture_axes_independent`, `L.error_independence_from_orthogonality`
`cor:dof-monotone`	Full	`L.lower_dof_lower_errors`
`cor:dof-ratio`	Full	`L.dof_ratio_predicts_error_ratio`
`cor:leverage-energy`	Full	`L.Physical.higher_leverage_same_caps_implies_lower_energy`, `L.Physical.positive_energy_gap_of_higher_leverage`
`cor:linear-approx`	Full	`L.bernoulli_justifies_linear_model`, `L.ordering_equivalence_exact`
`cor:physical-assumption-necessity`	Full	`L.Physical.positive_floor_requires_positive_joules_assumption`, `L.Physical.zero_model_energy_is_zero`

prop:dof-additive	Full	L.compose_dof, L.composition_dof_additive
thm:approx-bound	Full	L.linear_model_preserves_ordering, L.ordering_equivalence_exact
thm:composition	Full	L.composition_caps_additive, L.composition_dof_additive
thm:dof-reliability	Full	L.dof_reliability_isomorphism, L.isomorphism_preserves_failure_ordering
thm:england	Full	england_replication_inequality
thm:error-compound	Full	L.correctness_probability, L.system_is_correct
thm:error-independence	Full	L.error_independence_from_orthogonality
thm:error-prob	Full	L.error_probability_denom_pos, L.series_error_probability
thm:expected-errors	Full	L.expected_errors_from_linearity, L.expected_errors_linear
thm:five-way	Unmapped	*(no derived Lean handle found)*
thm:leverage-error	Full	L.leverage_error_tradeoff
thm:leverage-gap	Full	L.leverage_gap
thm:leverage-max	Full	L.leverage_caps_principle, L.leverage_maximization_principle
thm:metaprog	Full	L.metaprogramming_dominates, L.metaprogramming_unbounded_leverage
thm:mod-bound	Full	L.modification_eq_dof
thm:nominal-leverage	Full	L.Typing.capability_gap, L.Typing.nominal_dominates_duck
thm:optimal	Full	L.max_leverage_is_optimal, L.optimal_minimizes_error
thm:paper1-integration	Full	L.Typing.nominal_dominates_duck, L.Typing.paper1_is_leverage_instance
thm:paper2-integration	Full	L.SSOT.paper2_is_leverage_instance, L.SSOT.ssot_leverage_dominance
thm: physical-budget-boundary	Full	L.Physical.feasible_iff_floor_le_budget, L.Physical.infeasible_of_budget_lt_floor
thm:physical-energy-floor	Full	L.Physical.energyLowerBound_eq_joules_times_dof, L.Physical.lower_dof_lower_energy
thm:ssot-leverage	Full	L.SSOT.modification_ratio, L.SSOT.ssot_leverage_dominance
thm:testable-prediction	Full	L.testable_modification_prediction

Notes: (1) Full rows come from theorem-local inline anchors in this paper. (2) Derived rows are filled by dependency/scaffold claim-handle derivation (same paper-handle label across proof dependencies). (3) Unmapped means no local anchor and no derivable dependency support were found.
Auto summary: mapped 28/29 (full=28, derived=0, unmapped=1).

A.3 Lean Handle Map

Lean handle entry
AB1 DecisionQuotient.DecisionProblem.not_preservesOpt_iff_erasesDecisionRelevantDistinction
AB2 DecisionQuotient.DecisionProblem.surjective_abstraction_factors_or_erases
AB3 DecisionQuotient.DecisionProblem.collapseBeyondQuotient_physically_impossible
AB4 DecisionQuotient.DecisionProblem.surjective_abstraction_with_feasible_collapse_map_factors
AC1 DecisionQuotient.ClaimClosure.AtomicCircuitExports.AC1
AC3 DecisionQuotient.ClaimClosure.AtomicCircuitExports.AC3
AC4 DecisionQuotient.ClaimClosure.AtomicCircuitExports.AC4
AC5 DecisionQuotient.ClaimClosure.AtomicCircuitExports.AC5
AC6 DecisionQuotient.ClaimClosure.AtomicCircuitExports.AC6
AC8 DecisionQuotient.ClaimClosure.AtomicCircuitExports.AC8
AC9 DecisionQuotient.ClaimClosure.AtomicCircuitExports.AC9
AC11 DecisionQuotient.ClaimClosure.AtomicCircuitExports.AC11
ADV1 adversary_forces_n_minus_1_queries
ADV2 derivation_impossibility
ADV3 compile_macros_insufficient
ADV4 build_codegen_insufficient
ADV5 runtime_reflection_too_late

Lean handle entry (continued)
ALT1 protocol_is_concession
ALT2 protocol_not_alternative
AN1 DecisionQuotient.Physics.AssumptionNecessity.no_assumption_free_proof_of_refutable_claim
AN2 DecisionQuotient.Physics.AssumptionNecessity.countermodel_violates_some_assumption
AN3 DecisionQuotient.Physics.AssumptionNecessity.physical_claim_requires_physical_assumption
AN4 DecisionQuotient.Physics.AssumptionNecessity.physical_claim_requires_empirically_justified_physical_assumption
AN5 DecisionQuotient.Physics.AssumptionNecessity.strong_physical_no_go_meta
AQ1 DecisionQuotient.ClaimClosure.AQ1
AQ2 DecisionQuotient.ClaimClosure.AQ2
AQ3 DecisionQuotient.ClaimClosure.AQ3
AQ4 DecisionQuotient.ClaimClosure.AQ4
AQ5 DecisionQuotient.ClaimClosure.AQ5
AQ6 DecisionQuotient.ClaimClosure.AQ6
AQ7 DecisionQuotient.ClaimClosure.AQ7
AQ8 DecisionQuotient.ClaimClosure.AQ8
ARG1 PhysicalComplexity.AccessRegime.PhysicalDevice
ARG2 PhysicalComplexity.AccessRegime.AccessRegime
ARG3 PhysicalComplexity.AccessRegime.RegimeEval
ARG4 PhysicalComplexity.AccessRegime.RegimeSample
ARG5 PhysicalComplexity.AccessRegime.RegimeProof
ARG6 PhysicalComplexity.AccessRegime.RegimeWithCertificate
ARG7 PhysicalComplexity.AccessRegime.RegimeEvalOn
ARG8 PhysicalComplexity.AccessRegime.RegimeSampleOn
ARG9 PhysicalComplexity.AccessRegime.RegimeProofOn
ARG10 PhysicalComplexity.AccessRegime.RegimeWithCertificateOn
ARG11 PhysicalComplexity.AccessRegime.HardUnderEval
ARG12 PhysicalComplexity.AccessRegime.AuditableWithCertificate
ARG13 PhysicalComplexity.AccessRegime.certificate_upgrades_regime
ARG14 PhysicalComplexity.AccessRegime.certificate_upgrades_regime_on
ARG15 PhysicalComplexity.AccessRegime.physical_succinct_certification_hard
ARG16 PhysicalComplexity.AccessRegime.certificate_amortizes_hardness
ARG17 PhysicalComplexity.AccessRegime.regime_upgrade_with_certificate
ARG18 PhysicalComplexity.AccessRegime.regime_upgrade_with_certificate_on
ARG19 PhysicalComplexity.AccessRegime.AccessChannelLaw
ARG20 PhysicalComplexity.AccessRegime.FiveWayMeet
AXI1 shape_cannot_distinguish
AXM1 complete_mono
AXM2 completeD_mono
AXM3 minimal_no_redundant_axes
AXM4 semantically_minimal_implies_independent
BA1 DecisionQuotient.Physics.BoundedAcquisition.BoundedRegion
BA2 DecisionQuotient.Physics.BoundedAcquisition.acquisition_rate_bound
BA3 DecisionQuotient.Physics.BoundedAcquisition.acquisitions_are_transitions

Lean handle entry (continued)
BA4 DecisionQuotient.Physics.BoundedAcquisition.one_bit_per_transition
BA5 DecisionQuotient.Physics.BoundedAcquisition.resolution_reads_sufficient
BA6 DecisionQuotient.Physics.BoundedAcquisition.srank_le_resolution_bits
BA7 DecisionQuotient.Physics.BoundedAcquisition.energy_ge_srank_cost
BA8 DecisionQuotient.Physics.BoundedAcquisition.srank_one_energy_minimum
BA9 DecisionQuotient.Physics.BoundedAcquisition.physical_grounding_bundle
BA10 DecisionQuotient.Physics.BoundedAcquisition.counting_gap_theorem
BAS1 Ssot.correctness_forcing
BAS2 Ssot.dof_inconsistency_potential
BAS3 Ssot.dof_gt_one_inconsistent
BB1 DecisionQuotient.BayesianDQ
BB2 DecisionQuotient.BayesianDQ.certaintyGain
BB3 DecisionQuotient.dq_is_bayesian_certainty_fraction
BB4 DecisionQuotient.bayesian_dq_matches_physics_dq
BB5 DecisionQuotient.dq_derived_from_bayes
BC1 DecisionQuotient.Foundations.counting_nonneg
BC2 DecisionQuotient.Foundations.counting_total
BC3 DecisionQuotient.Foundations.counting_additive
BC4 DecisionQuotient.Foundations.bayes_from_conditional
BC5 DecisionQuotient.Foundations.entropy_contraction
BF1 DecisionQuotient.certainty_of_not_nondegenerateBelief
BF2 DecisionQuotient.nondegenerateBelief_of_uncertaintyForced
BF3 DecisionQuotient.forced_action_under_uncertainty
BF4 DecisionQuotient.bayes_update_exists_of_nondegenerateBelief
BND1 ssot_upper_bound
BND2 non_ssot_lower_bound
BND3 ssot_advantage_unbounded
BND4 ClaimClosure.arbitrary_reduction_factor
BR1 DecisionQuotient.Bridges.eth_structural_rank_exponential_hardness
BR2 DecisionQuotient.Bridges.fisher_rank_lower_bounds_sufficient_set
BR3 DecisionQuotient.Bridges.fpt_srank_parameterized_dichotomy
BR4 DecisionQuotient.Bridges.tur_srank_thermodynamic_cost
BR5 DecisionQuotient.Bridges.dichotomy_eth_complete_classification
BR6 DecisionQuotient.Bridges.reduction_eth_conp_exponential
BR7 DecisionQuotient.Bridges.geometry_covering_certificate_complexity
BR8 DecisionQuotient.Bridges.rate_distortion_fisher_information_bridge
BR9 DecisionQuotient.Bridges.counting_complexity_sharp_p_hardness
BR10 DecisionQuotient.Bridges.approximation_counting_hardness_bridge
BRG1 analysis_has_positive_ev
BRG2 ignorant_choice_has_cost
BRG3 retrofit_cost_dominates
CAP1 ClaimClosure.cap_encoding_conditional
CAP2 ssot_guarantees_coherence

Lean handle entry (continued)
CAP3 non_ssot_permits_incoherence
CC1 DecisionQuotient.ClaimClosure.RegimeSimulation
CC2 DecisionQuotient.ClaimClosure.adq_ordering
CC3 DecisionQuotient.ClaimClosure.system_transfer_licensed_iff_snapshot
CC4 DecisionQuotient.ClaimClosure.anchor_sigma2p_complete_conditional
CC5 DecisionQuotient.ClaimClosure.anchor_sigma2p_reduction_core
CC6 DecisionQuotient.ClaimClosure.anchor_query_relation_false_iff
CC7 DecisionQuotient.ClaimClosure.anchor_query_relation_true_iff
CC8 DecisionQuotient.ClaimClosure.boundaryCharacterized_iff_exists_sufficient_subset
CC9 DecisionQuotient.ClaimClosure.bounded_actions_detectable
CC10 DecisionQuotient.ClaimClosure.bridge_boundary_represented_family
CC11 DecisionQuotient.ClaimClosure.bridge_failure_witness_non_one_step
CC12 DecisionQuotient.ClaimClosure.bridge_transfer_iff_one_step_class
CC13 DecisionQuotient.ClaimClosure.certified_total_bits_split_core
CC14 DecisionQuotient.ClaimClosure.cost_asymmetry_eth_conditional
CC15 DecisionQuotient.ClaimClosure.declaredBudgetSlice
CC16 DecisionQuotient.ClaimClosure.declaredRegimeFamily_complete
CC17 DecisionQuotient.ClaimClosure.declared_physics_no_universal_exact_certifier_core
CC18 DecisionQuotient.ClaimClosure.dichotomy_conditional
CC19 DecisionQuotient.ClaimClosure.epsilon_admissible_iff_raw_lt_certified_total_core
CC20 DecisionQuotient.ClaimClosure.exact_admissible_iff_raw_lt_certified_total_core
CC21 DecisionQuotient.ClaimClosure.exact_certainty_inflation_under_hardness_core
CC22 DecisionQuotient.ClaimClosure.exact_raw_eq_certified_iff_certainty_inflation_core
CC23 DecisionQuotient.ClaimClosure.exact_raw_only_of_no_exact_admissible_core
CC24 DecisionQuotient.ClaimClosure.explicit_assumptions_required_of_not_excused_core
CC25 DecisionQuotient.ClaimClosure.explicit_state_upper_core
CC26 DecisionQuotient.ClaimClosure.hard_family_all_coords_core
CC27 DecisionQuotient.ClaimClosure.horizonTwoWitness_immediate_empty_sufficient
CC28 DecisionQuotient.ClaimClosure.horizon_gt_one_bridge_can_fail_on_sufficiency
CC29 DecisionQuotient.ClaimClosure.information_barrier_opt_oracle_core
CC30 DecisionQuotient.ClaimClosure.information_barrier_state_batch_core
CC31 DecisionQuotient.ClaimClosure.information_barrier_value_entry_core
CC32 DecisionQuotient.ClaimClosure.integrity_resource_bound_for_sufficiency
CC33 DecisionQuotient.ClaimClosure.integrity_universal_applicability_core
CC34 DecisionQuotient.ClaimClosure.meta_coordinate_irrelevant_of_invariance_on_declared_slice
CC35 DecisionQuotient.ClaimClosure.meta_coordinate_not_relevant_on_declared_slice
CC36 DecisionQuotient.ClaimClosure.minsuff_collapse_core
CC37 DecisionQuotient.ClaimClosure.minsuff_collapse_to_conp_conditional
CC38 DecisionQuotient.ClaimClosure.minsuff_conp_complete_conditional
CC39 DecisionQuotient.ClaimClosure.no_auto_minimize_of_p_neq_conp
CC40 DecisionQuotient.ClaimClosure.no_exact_claim_admissible_under_hardness_core
CC41 DecisionQuotient.ClaimClosure.no_exact_claim_under_declared_assumptions_unless_excused_core
CC42 DecisionQuotient.ClaimClosure.no_exact_identifier_implies_not_boundary_characterized

Lean handle entry (continued)
CC43 DecisionQuotient.ClaimClosure.no_uncertified_exact_claim_core
CC44 DecisionQuotient.ClaimClosure.one_step_bridge
CC45 DecisionQuotient.ClaimClosure.oracle_lattice_transfer_as_regime_simulation
CC46 DecisionQuotient.ClaimClosure.physical_crossover_above_cap_core
CC47 DecisionQuotient.ClaimClosure.physical_crossover_core
CC48 DecisionQuotient.ClaimClosure.physical_crossover_hardness_core
CC49 DecisionQuotient.ClaimClosure.physical_crossover_policy_core
CC50 DecisionQuotient.ClaimClosure.process_bridge_failure_witness
CC51 DecisionQuotient.ClaimClosure.poseAnchorQuery
CC52 DecisionQuotient.ClaimClosure.pose_returns_anchor_query_object
CC53 DecisionQuotient.ClaimClosure.posed_anchor_checked_true_implies_truth
CC54 DecisionQuotient.ClaimClosure.posed_anchor_exact_claim_admissible_iff_competent
CC55 DecisionQuotient.ClaimClosure.posed_anchor_exact_claim_requires_evidence
CC56 DecisionQuotient.ClaimClosure.posed_anchor_no_competence_no_exact_claim
CC57 DecisionQuotient.ClaimClosure.posed_anchor_query_truth_iff_exists_anchor
CC58 DecisionQuotient.ClaimClosure.posed_anchor_query_truth_iff_exists_forall
CC59 DecisionQuotient.ClaimClosure.posed_anchor_signal_positive_certified_implies_admissible
CC60 DecisionQuotient.ClaimClosure.query_obstruction_boolean_corollary
CC61 DecisionQuotient.ClaimClosure.query_obstruction_finite_state_core
CC62 DecisionQuotient.ClaimClosure.regime_core_claim_proved
CC63 DecisionQuotient.ClaimClosure.regime_simulation_transfers_hardness
CC64 DecisionQuotient.ClaimClosure.reusable_heuristic_of_detectable
CC65 DecisionQuotient.ClaimClosure.selectorSufficient_not_implies_setSufficient
CC66 DecisionQuotient.ClaimClosure.separable_detectable
CC67 DecisionQuotient.ClaimClosure.snapshot_vs_process_typed_boundary
CC68 DecisionQuotient.ClaimClosure.standard_assumption_ledger_unpack
CC69 DecisionQuotient.ClaimClosure.stochastic_objective_bridge_can_fail_on_sufficiency
CC70 DecisionQuotient.ClaimClosure.subproblem_hardness_lifts_to_full
CC71 DecisionQuotient.ClaimClosure.subproblem_transfer_as_regime_simulation
CC72 DecisionQuotient.ClaimClosure.sufficiency_conp_complete_conditional
CC73 DecisionQuotient.ClaimClosure.sufficiency_conp_reduction_core
CC74 DecisionQuotient.ClaimClosure.sufficiency_iff_dq_ratio
CC75 DecisionQuotient.ClaimClosure.sufficiency_iff_projectedOptCover_eq_opt
CC76 DecisionQuotient.ClaimClosure.thermo_conservation_additive_core
CC77 DecisionQuotient.ClaimClosure.thermo_energy_carbon_lift_core
CC78 DecisionQuotient.ClaimClosure.thermo_eventual_lift_core
CC79 DecisionQuotient.ClaimClosure.thermo_hardness_bundle_core
CC80 DecisionQuotient.ClaimClosure.thermo_mandatory_cost_core
CC81 DecisionQuotient.ClaimClosure.tractable_bounded_core
CC82 DecisionQuotient.ClaimClosure.tractable_separable_core
CC83 DecisionQuotient.ClaimClosure.tractable_subcases_conditional
CC84 DecisionQuotient.ClaimClosure.tractable_tree_core
CC85 DecisionQuotient.ClaimClosure.transition_coupled_bridge_can_fail_on_sufficiency

Lean handle entry (continued)
CC86 DecisionQuotient.ClaimClosure.tree_structure_detectable
CC87 DecisionQuotient.ClaimClosure.typed_claim_admissibility_core
CC88 DecisionQuotient.ClaimClosure.typed_static_class_completeness
CC89 DecisionQuotient.ClaimClosure.universal_solver_framing_core
CCC1 DecisionQuotient.CC.anchor_sigma2p_complete_conditional
CCC2 DecisionQuotient.CC.cost_asymmetry_eth_conditional
CCC3 DecisionQuotient.CC.dichotomy_conditional
CCC4 DecisionQuotient.CC.minsuff_collapse_to_conp_conditional
CCC5 DecisionQuotient.CC.minsuff_conp_complete_conditional
CCC6 DecisionQuotient.CC.sufficiency_conp_complete_conditional
CCC7 DecisionQuotient.CC.tractable_subcases_conditional
CF1 DecisionQuotient.Physics.ConstraintForcing.laws_not_determined_of_parameter_separation
CF2 DecisionQuotient.Physics.ConstraintForcing.logic_time_not_sufficient_for_unique_law
CF3 DecisionQuotient.Physics.ConstraintForcing.laws_determined_implies_objective_determined
CF4 DecisionQuotient.Physics.ConstraintForcing.objective_not_determined_of_parameter_separation
CF5 DecisionQuotient.Physics.ConstraintForcing.forcedDecisionBits_pos_of_deadline
CF6 DecisionQuotient.Physics.ConstraintForcing.actionForced_of_deadline
CF7 DecisionQuotient.Physics.ConstraintForcing.nondegenerateBelief_of_deadline_and_uncertainty
CF8 DecisionQuotient.Physics.ConstraintForcing.forced_decision_implies_positive_landauer_cost
CF9 DecisionQuotient.Physics.ConstraintForcing.forced_decision_implies_positive_nv_work
CH1 DecisionQuotient.ClaimClosure.CH1
CH2 DecisionQuotient.ClaimClosure.CH2
CH3 DecisionQuotient.ClaimClosure.CH3
CH5 DecisionQuotient.ClaimClosure.CH5
CH6 DecisionQuotient.ClaimClosure.CH6
CIA1 ClaimClosure.ClassicalInfoAssumptions
CMP1 complexity_gap_unbounded
COH1 preference_incoherent
COH2 AbstractClassSystem.hedging_incoherent
COH3 determinate_truth_forces_ssot
CPL1 cost_ratio_eq_dof
CR1 DecisionQuotient.ConfigReduction.config_sufficiency_iff_behavior_preserving
CT1 DecisionQuotient.Physics.ClaimTransport.PhysicalEncoding
CT2 DecisionQuotient.Physics.ClaimTransport.physical_claim_lifts_from_core
CT3 DecisionQuotient.Physics.ClaimTransport.physical_claim_lifts_from_core_conditional
CT4 DecisionQuotient.Physics.ClaimTransport.physical_counterexample_yields_core_counterexample
CT5 DecisionQuotient.Physics.ClaimTransport.physical_counterexample_invalidates_core_rule
CT6 DecisionQuotient.Physics.ClaimTransport.no_physical_counterexample_of_core_theorem
CT7 DecisionQuotient.Physics.ClaimTransport.LawGapInstance
CT8 DecisionQuotient.Physics.ClaimTransport.lawGapEncoding
CT9 DecisionQuotient.Physics.ClaimTransport.lawGapPhysicalClaim
CT10 DecisionQuotient.Physics.ClaimTransport.law_gap_physical_claim_holds
CT11 DecisionQuotient.Physics.ClaimTransport.no_law_gap_counterexample

Lean handle entry (continued)
CT12 DecisionQuotient.Physics.ClaimTransport.physical_bridge_bundle
CV1 DecisionQuotient.Physics.Conversation.RecurrentCircuit
CV2 DecisionQuotient.Physics.Conversation.CoupledConversation
CV3 DecisionQuotient.Physics.Conversation.JointState
CV4 DecisionQuotient.Physics.Conversation.tick_uses_shared_node
CV5 DecisionQuotient.Physics.Conversation.tick_shared_is_merged_emissions
CV6 DecisionQuotient.Physics.Conversation.channel_projection_eq_iff_quantized_eq
CV7 DecisionQuotient.Physics.Conversation.clamp_projection_eq_iff_same_clamped_bit
CV10 DecisionQuotient.Physics.Conversation.explanationGap_add_explanationBits
CV11 DecisionQuotient.Physics.Conversation.toClaimReport
CV12 DecisionQuotient.Physics.Conversation.abstain_iff_no_answer
CV13 DecisionQuotient.Physics.Conversation.yes_no_iff_exact_claim
CV14 DecisionQuotient.Physics.Conversation.toReportSignal_completion_defined_of_budget
CV15 DecisionQuotient.Physics.Conversation.toReportSignal_signal_consistent_zero_certified
CV16 DecisionQuotient.Physics.Conversation.abstain_report_can_carry_explanation
CV17 DecisionQuotient.Physics.Conversation.clampDecisionEvent_iff_bitOps_pos
CV18 DecisionQuotient.Physics.Conversation.clamp_event_implies_positive_energy
CV19 DecisionQuotient.Physics.Conversation.BinaryAnswer
CV20 DecisionQuotient.Physics.Conversation.ConversationReport
CV21 DecisionQuotient.Physics.Conversation.explanationGap
CV26 DecisionQuotient.Physics.Conversation.toClaimReport_ne_epsilon
CV27 DecisionQuotient.Physics.Conversation.toReportSignal
CV28 DecisionQuotient.Physics.Conversation.toReportSignal_declares_bound
DC1 DecisionQuotient.StochasticSequential.static_stochastic_strict_separation
DC2 DecisionQuotient.StochasticSequential.stochastic_sequential_strict_separation
DC3 DecisionQuotient.StochasticSequential.complexity_dichotomy_hierarchy
DC4 DecisionQuotient.StochasticSequential.regime_hierarchy
DC5 DecisionQuotient.StochasticSequential.coNP_subset_PP
DC6 DecisionQuotient.StochasticSequential.PP_subset_PSPACE
DC7 DecisionQuotient.StochasticSequential.coNP_subset_PSPACE
DC8 DecisionQuotient.StochasticSequential.static_to_coNP
DC9 DecisionQuotient.StochasticSequential.stochastic_to_PP
DC10 DecisionQuotient.StochasticSequential.sequential_to_PSPACE
DC11 DecisionQuotient.StochasticSequential.ClaimClosure.claim_six_subcases
DC12 DecisionQuotient.StochasticSequential.ClaimClosure.claim_hierarchy
DC13 DecisionQuotient.StochasticSequential.ClaimClosure.claim_tractable_subcases_to_P
DC14 DecisionQuotient.StochasticSequential.stochastic_dichotomy
DC15 DecisionQuotient.StochasticSequential.above_threshold_hard
DC16 DecisionQuotient.StochasticSequential.StochasticAnchorSufficient
DC17 DecisionQuotient.StochasticSequential.StochasticAnchorSufficiencyCheck
DC18 DecisionQuotient.StochasticSequential.stochastic_anchor_check_iff
DC19 DecisionQuotient.StochasticSequential.stochastic_anchor_sufficient_of_stochastic_sufficient
DC20 DecisionQuotient.StochasticSequential.SequentialAnchorSufficient

Lean handle entry (continued)
DC21 DecisionQuotient.StochasticSequential.SequentialAnchorSufficiencyCheck
DC22 DecisionQuotient.StochasticSequential.sequential_anchor_check_iff
DC23 DecisionQuotient.StochasticSequential.sequential_anchor_sufficient_of_sequential_sufficient
DC24 DecisionQuotient.StochasticSequential.StochasticAnchorCheckInstance
DC25 DecisionQuotient.StochasticSequential.reduceMAJSAT_correct_anchor_strict
DC26 DecisionQuotient.StochasticSequential.reduceMAJSAT_to_stochastic_anchor_reduction
DC27 DecisionQuotient.StochasticSequential.SequentialAnchorCheckInstance
DC28 DecisionQuotient.StochasticSequential.reduceTQBF_correct_anchor
DC29 DecisionQuotient.StochasticSequential.reduceTQBF_to_sequential_anchor_reduction
DC30 DecisionQuotient.StochasticSequential.StatePotential
DC31 DecisionQuotient.StochasticSequential.utilityFromPotentialDrop_le_iff_nextPotential_ge
DC32 DecisionQuotient.StochasticSequential.utility_from_action_state_potential
DC33 DecisionQuotient.StochasticSequential.stochasticExpectedUtility_eq_neg_expectedActionPotential
DC34 DecisionQuotient.StochasticSequential.stochasticExpectedUtility_le_iff_expectedActionPotential_ge
DC35 DecisionQuotient.StochasticSequential.landauerEnergyFloor_nonneg
DC36 DecisionQuotient.StochasticSequential.landauerEnergyFloor_mono_bits
DC37 DecisionQuotient.StochasticSequential.thermodynamicCost_eq_landauerEnergyFloorRoom_states
DE1 DecisionQuotient.ClaimClosure.DE1
DE2 DecisionQuotient.ClaimClosure.DE2
DE3 DecisionQuotient.ClaimClosure.DE3
DE4 DecisionQuotient.ClaimClosure.DE4
DER1 all_derived_from_source
DER2 ClaimClosure.derivation_preserves_coherence_core
DES1 ClaimClosure.design_necessity
DG1 DecisionQuotient.Outside
DG2 DecisionQuotient.anchoredSlice
DG3 DecisionQuotient.anchoredSliceEquivOutside
DG4 DecisionQuotient.card_outside_eq_sub
DG5 DecisionQuotient.card_anchoredSlice
DG6 DecisionQuotient.card_anchoredSlice_eq_pow_sub
DG7 DecisionQuotient.card_anchoredSlice_eq_uniform
DG8 DecisionQuotient.anchoredSlice_mul_fixed_eq_full
DG9 DecisionQuotient.constantBoolDP
DG10 DecisionQuotient.firstCoordDP
DG11 DecisionQuotient.constantBoolDP_opt
DG12 DecisionQuotient.firstCoordDP_opt
DG13 DecisionQuotient.constantBoolDP_empty_sufficient
DG14 DecisionQuotient.firstCoordDP_empty_not_sufficient
DG15 DecisionQuotient.boolHypercube_node_count
DG16 DecisionQuotient.node_count_does_not_determine_edge_geometry
DG17 DecisionQuotient.DecisionProblem.edgeOnComplement
DG18 DecisionQuotient.DecisionProblem.edgeOnComplement_iff_not_sufficient
DOM1 strict_dominance

Lean handle entry (continued)
DP1 DecisionQuotient.DecisionProblem.minimalSufficient_iff_relevant
DP2 DecisionQuotient.DecisionProblem.relevantSet_is_minimal
DP3 DecisionQuotient.DecisionProblem.sufficient_implies_selectorSufficient
DP4 DecisionQuotient.ClaimClosure.DecisionProblem.epsOpt_zero_eq_opt
DP5 DecisionQuotient.ClaimClosure.DecisionProblem.sufficient_iff_zeroEpsilonSufficient
DP6 DecisionQuotient.ClaimClosure.DP6
DP7 DecisionQuotient.ClaimClosure.DP7
DP8 DecisionQuotient.ClaimClosure.DP8
DQ1 DecisionQuotient.ClaimClosure.DQ1
DQ2 DecisionQuotient.ClaimClosure.DQ2
DQ3 DecisionQuotient.ClaimClosure.DQ3
DQ4 DecisionQuotient.ClaimClosure.DQ4
DQ5 DecisionQuotient.ClaimClosure.DQ5
DQ6 DecisionQuotient.ClaimClosure.DQ6
DQ7 DecisionQuotient.ClaimClosure.DQ7
DQ8 DecisionQuotient.ClaimClosure.DQ8
DS1 DecisionQuotient.ClaimClosure.DS1
DS2 DecisionQuotient.ClaimClosure.DS2
DS3 DecisionQuotient.ClaimClosure.DS3
DS4 DecisionQuotient.ClaimClosure.DS4
DS5 DecisionQuotient.ClaimClosure.DS5
DS6 DecisionQuotient.ClaimClosure.DS6
DT1 DecisionQuotient.Physics.DecisionTime.TimedState
DT2 DecisionQuotient.Physics.DecisionTime.DecisionProcess
DT3 DecisionQuotient.Physics.DecisionTime.tick
DT4 DecisionQuotient.Physics.DecisionTime.DecisionEvent
DT5 DecisionQuotient.Physics.DecisionTime.TimeUnitStep
DT6 DecisionQuotient.Physics.DecisionTime.time_is_discrete
DT7 DecisionQuotient.Physics.DecisionTime.time_coordinate_falsifiable
DT8 DecisionQuotient.Physics.DecisionTime.tick_increments_time
DT9 DecisionQuotient.Physics.DecisionTime.tick_decision_witness
DT10 DecisionQuotient.Physics.DecisionTime.tick_is_decision_event
DT11 DecisionQuotient.Physics.DecisionTime.decision_event_implies_time_unit
DT12 DecisionQuotient.Physics.DecisionTime.decision_taking_place_is_unit_of_time
DT13 DecisionQuotient.Physics.DecisionTime.decision_event_iff_eq_tick
DT14 DecisionQuotient.Physics.DecisionTime.run
DT15 DecisionQuotient.Physics.DecisionTime.run_time_exact
DT16 DecisionQuotient.Physics.DecisionTime.run_elapsed_time_eq_ticks
DT17 DecisionQuotient.Physics.DecisionTime.decisionTrace
DT18 DecisionQuotient.Physics.DecisionTime.decisionTrace_length_eq_ticks
DT19 DecisionQuotient.Physics.DecisionTime.decision_count_equals_elapsed_time
DT20 DecisionQuotient.Physics.DecisionTime.SubstrateKind
DT21 DecisionQuotient.Physics.DecisionTime.SubstrateModel

Lean handle entry (continued)
DT22 DecisionQuotient.Physics.DecisionTime.substrate_step_realizes_decision_event
DT23 DecisionQuotient.Physics.DecisionTime.substrate_step_is_time_unit
DT24 DecisionQuotient.Physics.DecisionTime.time_unit_law_substrate_invariant
DUC1 duck_localization_linear
EI1 DecisionQuotient.ThermodynamicLift.energy_ge_kbt_nat_entropy
EI2 DecisionQuotient.ThermodynamicLift.energy_ground_state_tracks_entropy
EI4 DecisionQuotient.ThermodynamicLift.landauerJoulesPerBit_pos
EI5 DecisionQuotient.ThermodynamicLift.neukart_vinokur_duality
EMB1 semantic_non_embeddability
ENC1 encoding_location_gap
ENT1 Entropy.ClassicalEntropyAssumptions
EP1 DecisionQuotient.Physics.LocalityPhysics.landauer_principle
EP2 DecisionQuotient.Physics.LocalityPhysics.finite_regional_energy
EP3 DecisionQuotient.Physics.LocalityPhysics.finite_signal_speed
EP4 DecisionQuotient.Physics.LocalityPhysics.nontrivial_physics
EXT1 extension_impossibility
FI3 DecisionQuotient.FunctionalInformation.functional_information_from_counting
FI6 DecisionQuotient.FunctionalInformation.functional_information_from_thermodynamics
FI7 DecisionQuotient.FunctionalInformation.first_principles_thermo_coincide
FLP1 ClaimClosure.static_flp_core
FN7 DecisionQuotient.BayesOptimalityProof.KL_nonneg
FN8 DecisionQuotient.BayesOptimalityProof.entropy_le_crossEntropy
FN12 DecisionQuotient.BayesOptimalityProof.crossEntropy_eq_entropy_add_KL
FN14 DecisionQuotient.BayesOptimalityProof.bayes_is_optimal
FN15 DecisionQuotient.BayesOptimalityProof.KL_eq_sum_klFun
FN16 DecisionQuotient.BayesOptimalityProof.KL_eq_zero_iff_eq
FORC2 typing_choice_unavoidable
FORC3 capability_foreclosure_irreversible
FP1 DecisionQuotient.Physics.LocalityPhysics.trivial_states_all_equal
FP2 DecisionQuotient.Physics.LocalityPhysics.equal_states_constant_function
FP3 DecisionQuotient.Physics.LocalityPhysics.constant_function_singleton_image
FP4 DecisionQuotient.Physics.LocalityPhysics.singleton_image_zero_entropy
FP5 DecisionQuotient.Physics.LocalityPhysics.zero_entropy_no_information
FP6 DecisionQuotient.Physics.LocalityPhysics.triviality_implies_no_information
FP7 DecisionQuotient.Physics.LocalityPhysics.information_requires_nontriviality
FP8 DecisionQuotient.Physics.LocalityPhysics.atypical_states_rare
FP9 DecisionQuotient.Physics.LocalityPhysics.random_misses_target
FP10 DecisionQuotient.Physics.LocalityPhysics.errors_accumulate
FP11 DecisionQuotient.Physics.LocalityPhysics.wrong_paths_dominate
FP12 DecisionQuotient.Physics.LocalityPhysics.second_law_from_counting
FP13 DecisionQuotient.Physics.LocalityPhysics.verification_is_information
FP14 DecisionQuotient.Physics.LocalityPhysics.entropy_is_information
FP15 DecisionQuotient.Physics.LocalityPhysics.landauer_structure

Lean handle entry (continued)
FPT1 DecisionQuotient.Physics.LocalityPhysics.Timeline
FPT2 DecisionQuotient.Physics.LocalityPhysics.FPT2_function_deterministic
FPT3 DecisionQuotient.Physics.LocalityPhysics.FPT3_outputs_differ_inputs_differ
FPT4 DecisionQuotient.Physics.LocalityPhysics.FPT4_step_requires_distinct_moments
FPT5 DecisionQuotient.Physics.LocalityPhysics.FPT5_distinct_moments_positive_duration
FPT6 DecisionQuotient.Physics.LocalityPhysics.FPT6_step_takes_positive_time
FPT7 DecisionQuotient.Physics.LocalityPhysics.FPT7_no_instantaneous_steps
FPT8 DecisionQuotient.Physics.LocalityPhysics.FPT8_propagation_takes_time
FPT9 DecisionQuotient.Physics.LocalityPhysics.FPT9_speed_bounded_by_positive_time
FPT10 DecisionQuotient.Physics.LocalityPhysics.FPT10_ec3_is_logical
FS1 DecisionQuotient.Statistics.sum_fisherScore_eq_srank
FS2 DecisionQuotient.Statistics.fisherMatrix_rank_eq_srank
FS3 DecisionQuotient.Statistics.fisherMatrix_trace_eq_srank
FS4 DecisionQuotient.Statistics.fisherScore_relevant
FS5 DecisionQuotient.Statistics.fisherScore_irrelevant
GE1 DecisionQuotient.ClaimClosure.GE1
GE2 DecisionQuotient.ClaimClosure.GE2
GE3 DecisionQuotient.ClaimClosure.GE3
GE4 DecisionQuotient.ClaimClosure.GE4
GE5 DecisionQuotient.ClaimClosure.GE5
GE6 DecisionQuotient.ClaimClosure.GE6
GE7 DecisionQuotient.ClaimClosure.GE7
GE8 DecisionQuotient.ClaimClosure.GE8
GE9 DecisionQuotient.ClaimClosure.GE9
GEN1 generated_file_is_second_encoding
GN1 DecisionQuotient.LogicGraph.isCycle
GN2 DecisionQuotient.LogicGraph.cycleWitnessBits_pos
GN3 DecisionQuotient.LogicGraph.pathProb_nonneg
GN4 DecisionQuotient.LogicGraph.pathSurprisal_nonneg_of_positive_mass
GN5 DecisionQuotient.LogicGraph.nontrivialityScore_unknown
GN6 DecisionQuotient.LogicGraph.observerEntropy_nonneg
GN7 DecisionQuotient.LogicGraph.dqFromEntropy_in_unit_interval
GN8 DecisionQuotient.LogicGraph.path_belief_forced_under_uncertainty
GN9 DecisionQuotient.LogicGraph.bayes_update_exists_for_observer_paths
GN10 DecisionQuotient.LogicGraph.cycle_witness_implies_positive_landauer
GN11 DecisionQuotient.LogicGraph.cycle_witness_implies_positive_nv_work
GN12 DecisionQuotient.LogicGraph.dna_erasure_implies_positive_landauer
GN13 DecisionQuotient.LogicGraph.dna_room_temp_environmental_stability
HD1 DecisionQuotient.HardnessDistribution.centralization_dominance_bundle
HD2 DecisionQuotient.HardnessDistribution.centralization_step_saves_n_minus_one
HD3 DecisionQuotient.HardnessDistribution.centralized_higher_leverage
HD4 DecisionQuotient.HardnessDistribution.complete_model_dominates_after_threshold
HD5 DecisionQuotient.HardnessDistribution.gap_conservation_card

Lean handle entry (continued)
HD6 DecisionQuotient.HardnessDistribution.generalizedTotal_with_saturation_eventually_constant
HD7 DecisionQuotient.HardnessDistribution.generalized_dominance_can_fail_without_right_boundedness
HD8 DecisionQuotient.HardnessDistribution.generalized_dominance_can_fail_without_wrong_growth
HD9 DecisionQuotient.HardnessDistribution.generalized_right_dominates_wrong_of_bounded_vs_identity_lower
HD10 DecisionQuotient.HardnessDistribution.generalized_right_eventually_dominates_wrong
HD11 DecisionQuotient.HardnessDistribution.hardnessEfficiency_eq_central_share
HD12 DecisionQuotient.HardnessDistribution.isRightHardness
HD13 DecisionQuotient.HardnessDistribution.isWrongHardness
HD14 DecisionQuotient.HardnessDistribution.linear_lt_exponential_plus_constant_eventually
HD15 DecisionQuotient.HardnessDistribution.native_dominates_manual
HD16 DecisionQuotient.HardnessDistribution.no_positive_slope_linear_represents_saturating
HD17 DecisionQuotient.HardnessDistribution.requiredWork
HD18 DecisionQuotient.HardnessDistribution.requiredWork_eq_affine_in_sites
HD19 DecisionQuotient.HardnessDistribution.right_dominates_wrong
HD20 DecisionQuotient.HardnessDistribution.saturatingSiteCost_eventually_constant
HD21 DecisionQuotient.HardnessDistribution.simplicityTax_grows
HD22 DecisionQuotient.HardnessDistribution.hardnessLowerBound
HD23 DecisionQuotient.HardnessDistribution.hardness_is_irreducible_required_work
HD24 DecisionQuotient.HardnessDistribution.totalDOF_eventually_constant_iff_zero_distributed
HD25 DecisionQuotient.HardnessDistribution.totalDOF_ge_intrinsic
HD26 DecisionQuotient.HardnessDistribution.totalExternalWork_eq_n_mul_gapCard
HD27 DecisionQuotient.HardnessDistribution.workGrowthDegree
HD28 DecisionQuotient.HardnessDistribution.workGrowthDegree_zero_iff_eventually_constant
HS1 DecisionQuotient.Physics.HeisenbergStrong.NoisyPhysicalEncoding
HS2 DecisionQuotient.Physics.HeisenbergStrong.HeisenbergStrongBinding
HS3 DecisionQuotient.Physics.HeisenbergStrong.strong_binding_implies_core_nontrivial
HS4 DecisionQuotient.Physics.HeisenbergStrong.strong_binding_yields_core_encoding_witness
HS5 DecisionQuotient.Physics.HeisenbergStrong.strong_binding_implies_physical_nontrivial_opt_assumption
HS6 DecisionQuotient.Physics.HeisenbergStrong.strong_binding_implies_nontrivial_opt_via_uncertainty
IA1 DecisionQuotient.ClaimClosure.IA1
IA2 DecisionQuotient.ClaimClosure.IA2
IA3 DecisionQuotient.ClaimClosure.IA3
IA4 DecisionQuotient.ClaimClosure.IA4
IA5 DecisionQuotient.ClaimClosure.IA5
IA6 DecisionQuotient.ClaimClosure.IA6
IA7 DecisionQuotient.ClaimClosure.IA7
IA8 DecisionQuotient.ClaimClosure.IA8
IA9 DecisionQuotient.ClaimClosure.IA9
IA10 DecisionQuotient.ClaimClosure.IA10
IA11 DecisionQuotient.ClaimClosure.IA11
IA12 DecisionQuotient.ClaimClosure.IA12
IA13 DecisionQuotient.ClaimClosure.IA13
IC1 DecisionQuotient.IntegrityCompetence.CertaintyInflation

Lean handle entry (continued)
IC2 DecisionQuotient.IntegrityCompetence.CompletionFractionDefined
IC3 DecisionQuotient.IntegrityCompetence.EvidenceForReport
IC4 DecisionQuotient.IntegrityCompetence.ExactCertaintyInflation
IC5 DecisionQuotient.IntegrityCompetence.Percent
IC6 DecisionQuotient.IntegrityCompetence.RLFFWeights
IC7 DecisionQuotient.IntegrityCompetence.ReportSignal
IC8 DecisionQuotient.IntegrityCompetence.ReportBitModel
IC9 DecisionQuotient.IntegrityCompetence.SignalConsistent
IC10 DecisionQuotient.IntegrityCompetence.admissible_irrational_strictly_more_than_rational
IC11 DecisionQuotient.IntegrityCompetence.admissible_matrix_counts
IC12 DecisionQuotient.IntegrityCompetence.abstain_signal_exists_with_guess_self
IC13 DecisionQuotient.IntegrityCompetence.certaintyInflation_iff_not_admissible
IC14 DecisionQuotient.IntegrityCompetence.certificationOverheadBits
IC15 DecisionQuotient.IntegrityCompetence.certificationOverheadBits_of_evidence
IC16 DecisionQuotient.IntegrityCompetence.certificationOverheadBits_of_no_evidence
IC17 DecisionQuotient.IntegrityCompetence.certifiedTotalBits
IC18 DecisionQuotient.IntegrityCompetence.certifiedTotalBits_ge_raw
IC19 DecisionQuotient.IntegrityCompetence.certifiedTotalBits_gt_raw_of_evidence
IC20 DecisionQuotient.IntegrityCompetence.certifiedTotalBits_of_evidence
IC21 DecisionQuotient.IntegrityCompetence.certifiedTotalBits_of_no_evidence
IC22 DecisionQuotient.IntegrityCompetence.claim_admissible_of_evidence
IC23 DecisionQuotient.IntegrityCompetence.competence_implies_integrity
IC24 DecisionQuotient.IntegrityCompetence.completion_fraction_defined_of_declared_bound
IC25 DecisionQuotient.IntegrityCompetence.epsilon_competence_implies_integrity
IC26 DecisionQuotient.IntegrityCompetence.evidence_nonempty_iff_claim_admissible
IC27 DecisionQuotient.IntegrityCompetence.evidence_of_claim_admissible
IC28 DecisionQuotient.IntegrityCompetence.exact_claim_admissible_iff_exact_evidence_nonempty
IC29 DecisionQuotient.IntegrityCompetence.exact_claim_requires_evidence
IC30 DecisionQuotient.IntegrityCompetence.exactCertaintyInflation_iff_no_exact_competence
IC31 DecisionQuotient.IntegrityCompetence.exact_raw_only_of_no_exact_admissible
IC32 DecisionQuotient.IntegrityCompetence.integrity_forces_abstention
IC33 DecisionQuotient.IntegrityCompetence.integrity_not_competent_of_nonempty_scope
IC34 DecisionQuotient.IntegrityCompetence.integrity_resource_bound
IC35 DecisionQuotient.IntegrityCompetence.no_completion_fraction_without_declared_bound
IC36 DecisionQuotient.IntegrityCompetence.overModelVerdict_rational_iff
IC37 DecisionQuotient.IntegrityCompetence.percentZero
IC38 DecisionQuotient.IntegrityCompetence.rlffBaseReward
IC39 DecisionQuotient.IntegrityCompetence.rlffReward
IC40 DecisionQuotient.IntegrityCompetence.rlff_abstain_strictly_prefers_no_certificates
IC41 DecisionQuotient.IntegrityCompetence.rlff_maximizer_has_evidence
IC42 DecisionQuotient.IntegrityCompetence.rlff_maximizer_is_admissible
IC43 DecisionQuotient.IntegrityCompetence.self_reflected_confidence_not_certification
IC44 DecisionQuotient.IntegrityCompetence.signal_certified_positive_implies_admissible

Lean handle entry (continued)
IC45 DecisionQuotient.IntegrityCompetence.signal_consistent_of_claim_admissible
IC46 DecisionQuotient.IntegrityCompetence.signal_no_evidence_forces_zero_certified
IC47 DecisionQuotient.IntegrityCompetence.signal_exact_no_competence_forces_zero_certified
IC48 DecisionQuotient.IntegrityCompetence.steps_run_scalar_always_defined
IC49 DecisionQuotient.IntegrityCompetence.steps_run_scalar_falsifiable
IC50 DecisionQuotient.IntegrityCompetence.zero_epsilon_competence_iff_exact
IE1 DecisionQuotient.ClaimClosure.IE1
IE2 DecisionQuotient.ClaimClosure.IE2
IE3 DecisionQuotient.ClaimClosure.IE3
IE4 DecisionQuotient.ClaimClosure.IE4
IE5 DecisionQuotient.ClaimClosure.IE5
IE6 DecisionQuotient.ClaimClosure.IE6
IE7 DecisionQuotient.ClaimClosure.IE7
IE8 DecisionQuotient.ClaimClosure.IE8
IE9 DecisionQuotient.ClaimClosure.IE9
IE10 DecisionQuotient.ClaimClosure.IE10
IE11 DecisionQuotient.ClaimClosure.IE11
IE12 DecisionQuotient.ClaimClosure.IE12
IE13 DecisionQuotient.ClaimClosure.IE13
IE14 DecisionQuotient.ClaimClosure.IE14
IE15 DecisionQuotient.ClaimClosure.IE15
IE16 DecisionQuotient.ClaimClosure.IE16
IE17 DecisionQuotient.ClaimClosure.IE17
IMP1 shape_provenance_impossible
IMP2 java_forced_to_composition
IN1 DecisionQuotient.Physics.Instantiation.Geometry
IN2 DecisionQuotient.Physics.Instantiation.Dynamics
IN3 DecisionQuotient.Physics.Instantiation.Circuit
IN4 DecisionQuotient.Physics.Instantiation.geometry_plus_dynamics_is_circuit
IN5 DecisionQuotient.Physics.Instantiation.DecisionInterpretation
IN6 DecisionQuotient.Physics.Instantiation.DecisionCircuit
IN7 DecisionQuotient.Physics.Instantiation.Molecule
IN8 DecisionQuotient.Physics.Instantiation.Reaction
IN9 DecisionQuotient.Physics.Instantiation.ReactionOutcome
IN10 DecisionQuotient.Physics.Instantiation.MoleculeGeometry
IN11 DecisionQuotient.Physics.Instantiation.MoleculeDynamics
IN12 DecisionQuotient.Physics.Instantiation.MoleculeCircuit
IN13 DecisionQuotient.Physics.Instantiation.MoleculeAsCircuit
IN14 DecisionQuotient.Physics.Instantiation.MoleculeAsDecisionCircuit
IN15 DecisionQuotient.Physics.Instantiation.molecule_decision_preserves_geometry
IN16 DecisionQuotient.Physics.Instantiation.molecule_decision_preserves_dynamics
IN17 DecisionQuotient.Physics.Instantiation.asDecisionCircuit
IN18 DecisionQuotient.Physics.Instantiation.asDecisionCircuit_preserves_circuit

Lean handle entry (continued)
IN19 DecisionQuotient.Physics.Instantiation.Configuration
IN20 DecisionQuotient.Physics.Instantiation.EnergyLandscape
IN21 DecisionQuotient.Physics.Instantiation.k_Boltzmann
IN22 DecisionQuotient.Physics.Instantiation.LandauerBound
IN23 DecisionQuotient.Physics.Instantiation.law_objective_schema
IN24 DecisionQuotient.Physics.Instantiation.law_opt_eq_feasible_of_gap
IN25 DecisionQuotient.Physics.Instantiation.law_opt_singleton_of_deterministic
IND1 both_requirements_independent
IND2 both_requirements_independent’
IND3 external_tools_not_derivation
IND4 language_semantics_is_derivation
IND5 ide_refactoring_not_derivation
INS1 Inconsistency.ssot_is_unique_optimum
INS2 Inconsistency.ssot_required
INS3 Inconsistency.ssot_unique_satisfier
INS4 Inconsistency.resolution_requires_external_choice
IP1 DecisionQuotient.Physics.LocalityPhysics.ec1_can_be_true
IP2 DecisionQuotient.Physics.LocalityPhysics.ec1_independent_of_ec2_ec3
IP3 DecisionQuotient.Physics.LocalityPhysics.ec2_can_be_true
IP4 DecisionQuotient.Physics.LocalityPhysics.ec2_independent_of_ec1_ec3
IP5 DecisionQuotient.Physics.LocalityPhysics.ec3_can_be_true
IP6 DecisionQuotient.Physics.LocalityPhysics.ec3_independent_of_ec1_ec2
IP7 DecisionQuotient.Physics.LocalityPhysics.empirical_claims_mutually_independent
IT1 DecisionQuotient.DecisionProblem.numOptClasses
IT3 DecisionQuotient.quotientEntropy_le_srank_binary
IT4 DecisionQuotient.numOptClasses_le_pow_srank_binary
IT5 DecisionQuotient.nontrivial_bounds_binary
IT6 DecisionQuotient.nontrivial_implies_srank_pos
IV1 DecisionQuotient.InteriorVerification.GoalClass
IV2 DecisionQuotient.InteriorVerification.InteriorDominanceVerifiable
IV3 DecisionQuotient.InteriorVerification.TautologicalSetIdentifiable
IV4 DecisionQuotient.InteriorVerification.agreeOnSet
IV5 DecisionQuotient.InteriorVerification.interiorParetoDominates
IV6 DecisionQuotient.InteriorVerification.interior_certificate_implies_non_rejection
IV7 DecisionQuotient.InteriorVerification.interior_dominance_implies_universal_non_rejection
IV8 DecisionQuotient.InteriorVerification.interior_dominance_not_full_sufficiency
IV9 DecisionQuotient.InteriorVerification.interior_verification_tractable_certificate
IV10 DecisionQuotient.InteriorVerification.not_sufficientOnSet_of_counterexample
IV11 DecisionQuotient.InteriorVerification.singleton_coordinate_interior_certificate
L1 Leverage.Physical.energyLowerBound_eq_joules_times_dof
L2 Leverage.Physical.feasible_iff_floor_le_budget
L3 Leverage.Physical.higher_leverage_same_caps_implies_lower_energy
L4 Leverage.Physical.infeasible_of_budget_lt_floor

Lean handle entry (continued)
L5 Leverage.Physical.lower_dof_lower_energy
L6 Leverage.Physical.positive_energy_gap_of_higher_leverage
L7 Leverage.Physical.positive_floor_requires_positive_joules_assumption
L8 Leverage.Physical.zero_model_energy_is_zero
L9 Leverage.SSOT.modification_ratio
L10 Leverage.SSOT.paper2_is_leverage_instance
L11 Leverage.SSOT.ssot_leverage_dominance
L12 Leverage.Typing.capability_gap
L13 Leverage.Typing.nominal_dominates_duck
L14 Leverage.Typing.paper1_is_leverage_instance
L15 Leverage.architecture_axes_independent
L16 Leverage.bernoulli_justifies_linear_model
L17 Leverage.compose_dof
L18 Leverage.composition_caps_additive
L19 Leverage.composition_dof_additive
L20 Leverage.correctness_probability
L21 Leverage.dof_ratio_predicts_error_ratio
L22 Leverage.dof_reliability_isomorphism
L23 Leverage.error_independence_from_orthogonality
L24 Leverage.error_probability_denom_pos
L25 Leverage.expected_errors_from_linearity
L26 Leverage.expected_errors_linear
L27 Leverage.isomorphism_preserves_failure_ordering
L28 Leverage.leverage_caps_principle
L29 Leverage.leverage_error_tradeoff
L30 Leverage.leverage_gap
L31 Leverage.leverage_maximization_principle
L32 Leverage.linear_model_preserves_ordering
L33 Leverage.lower_dof_lower_errors
L34 Leverage.max_leverage_is_optimal
L35 Leverage.metaprogramming_dominates
L36 Leverage.metaprogramming_unbounded_leverage
L37 Leverage.modification_eq_dof
L38 Leverage.optimal_minimizes_error
L39 Leverage.ordering_equivalence_exact
L40 Leverage.series_error_probability
L41 Leverage.system_is_correct
L42 Leverage.testable_modification_prediction
L43 dof_eq_srank
L44 dof_one_iff_max_leverage
L45 england_replication_inequality
L46 incoherent_srank_gt_one
L47 max_coherence_forces_tractability

Lean handle entry (continued)
L48 max_leverage_forces_dof_one
L49 srank_energy_lower_bound
L50 ssot_max_leverage
L51 ssot_srank_one
L52 succ_le_two_pow
L53 sufficiency_conp_hard
L54 thermodynamic_selection
L55 thermodynamic_selection_unconditional
L56 tractable_bounded_core
LNG1 ClaimClosure.language_realizability_criterion
LP1 DecisionQuotient.Physics.LocalityPhysics.SpacetimePoint
LP2 DecisionQuotient.Physics.LocalityPhysics.lightCone
LP3 DecisionQuotient.Physics.LocalityPhysics.causalPast
LP4 DecisionQuotient.Physics.LocalityPhysics.self_in_lightCone
LP5 DecisionQuotient.Physics.LocalityPhysics.self_in_causalPast
LP6 DecisionQuotient.Physics.LocalityPhysics.LocalRegion
LP7 DecisionQuotient.Physics.LocalityPhysics.canObserve
LP8 DecisionQuotient.Physics.LocalityPhysics.spacelikeSeparated
LP9 DecisionQuotient.Physics.LocalityPhysics.spacelike_disjoint_observation
LP11 DecisionQuotient.Physics.LocalityPhysics.LocalConfiguration
LP12 DecisionQuotient.Physics.LocalityPhysics.isLocallyValid
LP13 DecisionQuotient.Physics.LocalityPhysics.MergeResult
LP14 DecisionQuotient.Physics.LocalityPhysics.boardMerge
LP15 DecisionQuotient.Physics.LocalityPhysics.independent_configs_can_disagree
LP16 DecisionQuotient.Physics.LocalityPhysics.merge_compatible_iff
LP17 DecisionQuotient.Physics.LocalityPhysics.merge_contradiction_iff
LP18 DecisionQuotient.Physics.LocalityPhysics.locality_implies_possible_contradiction
LP19 DecisionQuotient.Physics.LocalityPhysics.Superposition
LP20 DecisionQuotient.Physics.LocalityPhysics.superposition_can_contain_contradictions
LP21 DecisionQuotient.Physics.LocalityPhysics.superposition_requires_separation
LP22 DecisionQuotient.Physics.LocalityPhysics.bell_separation_is_real
LP23 DecisionQuotient.Physics.LocalityPhysics.measurement_is_merge_contradiction
LP24 DecisionQuotient.Physics.LocalityPhysics.entanglement_is_shared_origin
LP31 DecisionQuotient.Physics.LocalityPhysics.complete_knowledge_requires_all_queries
LP32 DecisionQuotient.Physics.LocalityPhysics.finite_energy_constraint
LP33 DecisionQuotient.Physics.LocalityPhysics.self_knowledge_impossible_if_insufficient_energy
LP34 DecisionQuotient.Physics.LocalityPhysics.bounded_energy_forces_locality
LP35 DecisionQuotient.Physics.LocalityPhysics.locality_implies_independent_regions
LP36 DecisionQuotient.Physics.LocalityPhysics.independent_regions_imply_possible_contradiction
LP38 DecisionQuotient.Physics.LocalityPhysics.pne_np_necessary_for_physics
LP39 DecisionQuotient.Physics.LocalityPhysics.matter_exists_because_pne_np
LP40 DecisionQuotient.Physics.LocalityPhysics.physics_is_the_game
LP41 DecisionQuotient.Physics.LocalityPhysics.without_positive_query_cost_no_bound

Lean handle entry (continued)

LP42 DecisionQuotient.Physics.LocalityPhysics.without_nontrivial_states_no_disagreement

LP43 DecisionQuotient.Physics.LocalityPhysics.without_separation_no_independence

LP44 DecisionQuotient.Physics.LocalityPhysics.without_finite_capacity_no_gap

LP45 DecisionQuotient.Physics.LocalityPhysics.all_premises_used

LP46 DecisionQuotient.Physics.LocalityPhysics.premises_necessary_and_sufficient

LP50 DecisionQuotient.Physics.LocalityPhysics.shannon_value_is_intractability

LP51 DecisionQuotient.Physics.LocalityPhysics.economic_value_requires_scarcity

LP52 DecisionQuotient.Physics.LocalityPhysics.thermodynamic_value_requires_gradient

LP53 DecisionQuotient.Physics.LocalityPhysics.voi_requires_uncertainty

LP54 DecisionQuotient.Physics.LocalityPhysics.physics_requires_intractability

LP55 DecisionQuotient.Physics.LocalityPhysics.value_is_intractability

LP56 DecisionQuotient.Physics.LocalityPhysics.observers_value_the_intractable

LP57 DecisionQuotient.Physics.LocalityPhysics.finite_steps_finite_coverage

LP58 DecisionQuotient.Physics.LocalityPhysics.counting_gap

LP59 DecisionQuotient.Physics.LocalityPhysics.time_is_counting

LP60 DecisionQuotient.Physics.LocalityPhysics.gap_equivalence

LP61 DecisionQuotient.Physics.LocalityPhysics.light_cone_is_time_gap

MDL1 model_completeness

MDL2 mechanism_exhaustiveness

MDL3 only_at_definition_works

MDL4 mechanism_structural_capability

MI1 DecisionQuotient.ClaimClosure.MI1

MI2 DecisionQuotient.ClaimClosure.MI2

MI3 DecisionQuotient.ClaimClosure.MI3

MI4 DecisionQuotient.ClaimClosure.MI4

MI5 DecisionQuotient.ClaimClosure.MI5

MN1 DecisionQuotient.Physics.MeasureNecessity.quantitative_claim_has_measure

MN2 DecisionQuotient.Physics.MeasureNecessity.stochastic_claim_has_probability_measure

MN3 DecisionQuotient.Physics.MeasureNecessity.stochastic_claim_has_measure

MN4 DecisionQuotient.Physics.MeasureNecessity.count_univ_bool

MN5 DecisionQuotient.Physics.MeasureNecessity.counting_measure_not_probability_on_bool

MN6 DecisionQuotient.Physics.MeasureNecessity.deterministic_dirac_is_probability

MN7 DecisionQuotient.Physics.MeasureNecessity.quantitative_value_depends_on_measure

MN8 DecisionQuotient.Physics.MeasureNecessity.deterministic_models_still_measure_based

MN9 DecisionQuotient.Physics.MeasureNecessity.measure_does_not_imply_probability

MN10 DecisionQuotient.Physics.MeasureNecessity.quantitative_measure_is_logical_prerequisite

MN11 DecisionQuotient.Physics.MeasureNecessity.stochastic_probability_is_logical_prerequisite

NOM1 nominal_centralized

NOM2 nominal_localization_constant_semantic

OR1 DecisionQuotient.Physics.ObserverRelativeState.ObserverClass

OR2 DecisionQuotient.Physics.ObserverRelativeState.obsEquiv

OR3 DecisionQuotient.Physics.ObserverRelativeState.EffectiveStateSpace

OR4 DecisionQuotient.Physics.ObserverRelativeState.project_eq_iff

Lean handle entry (continued)
OR5 DecisionQuotient.Physics.ObserverRelativeState.observer_relative_equivalence_witness
OR6 DecisionQuotient.Physics.ObserverRelativeState.PhysicalObserverClass
OR7 DecisionQuotient.Physics.ObserverRelativeState.PhysicalStateSpace
OR8 DecisionQuotient.Physics.ObserverRelativeState.physical_state_space_has_instance_witness
OR9 DecisionQuotient.Physics.ObserverRelativeState.physical_observer_relative_effective_space
ORA1 oracle_arbitrary
PA1 DecisionQuotient.Physics.AnchorChecks.obsEquiv_all_of_effective_subsingleton
PA2 DecisionQuotient.Physics.AnchorChecks.stochasticAnchorSufficient_iff_exists_anchor_singleton
PA3 DecisionQuotient.Physics.AnchorChecks.stochastic_anchor_check_iff_exists_anchor_singleton
PA4 DecisionQuotient.Physics.AnchorChecks.stochastic_sufficient_of_observer_collapse_and_seed
PA5 DecisionQuotient.Physics.AnchorChecks.stochastic_anchor_check_of_observer_collapse_and_seed
PA6 DecisionQuotient.Physics.AnchorChecks.sequential_sufficient_of_observer_collapse
PA7 DecisionQuotient.Physics.AnchorChecks.sequential_anchor_check_of_observer_collapse
PA8 DecisionQuotient.Physics.AnchorChecks.physical_observer_collapse_implies_obsEquiv_all
PA9 DecisionQuotient.Physics.AnchorChecks.physical_stochastic_anchor_check_of_observer_collapse_and_seed
PBC1 DecisionQuotient.PhysicalBudgetCrossover.CrossoverAt
PBC2 DecisionQuotient.PhysicalBudgetCrossover.SuccinctInfeasible
PBC3 DecisionQuotient.PhysicalBudgetCrossover.SuccinctUnbounded
PBC4 DecisionQuotient.PhysicalBudgetCrossover.explicit_infeasible_succinct_feasible_of_crossover
PBC5 DecisionQuotient.PhysicalBudgetCrossover.exists_least_crossover_point
PBC6 DecisionQuotient.PhysicalBudgetCrossover.has_crossover_of_bounded_succinct_unbounded_explicit
PBC7 DecisionQuotient.PhysicalBudgetCrossover.explicit_eventual_infeasibility_of_monotone_and_witness
PBC8 DecisionQuotient.PhysicalBudgetCrossover.crossover_eventually_of_eventual_split
PBC9 DecisionQuotient.PhysicalBudgetCrossover.payoff_threshold_explicit_vs_succinct
PBC10 DecisionQuotient.PhysicalBudgetCrossover.no_universal_survivor_without_succinct_bound
PBC11 DecisionQuotient.PhysicalBudgetCrossover.policy_closure_at_divergence
PBC12 DecisionQuotient.PhysicalBudgetCrossover.policy_closure_beyond_divergence
PH1 PhysicalComplexity.k_Boltzmann
PH2 PhysicalComplexity.PhysicalComputer
PH3 PhysicalComplexity.bit_energy_cost
PH4 PhysicalComplexity.Landauer_bound
PH5 PhysicalComplexity.InstanceSize
PH6 PhysicalComplexity.ComputationalRequirement
PH7 PhysicalComplexity.coNP_requirement
PH8 PhysicalComplexity.coNP_physically_impossible
PH9 PhysicalComplexity.coNP_not_in_P_physically
PH10 PhysicalComplexity.sufficiency_physically_impossible
PH11 DecisionQuotient.PhysicalComplexity.PhysicalCollapseAtRequirement
PH12 DecisionQuotient.PhysicalComplexity.no_physical_collapse_at_requirement
PH13 DecisionQuotient.PhysicalComplexity.canonical_physical_collapse_impossible
PH14 DecisionQuotient.PhysicalComplexity.p_eq_np_physically_impossible_of_collapse_map
PH15 DecisionQuotient.PhysicalComplexity.p_eq_np_physically_impossible_canonical
PH16 DecisionQuotient.PhysicalComplexity.P_eq_NP_via_SAT

Lean handle entry (continued)
PH17 DecisionQuotient.PhysicalComplexity.SAT3ReductionBridge
PH18 DecisionQuotient.PhysicalComplexity.sat_reduction_transfers_energy_lower_bound
PH19 DecisionQuotient.PhysicalComplexity.physical_collapse_of_polytime_sat_realization
PH20 DecisionQuotient.PhysicalComplexity.p_eq_np_physically_impossible_via_sat_bridge
PH21 DecisionQuotient.PhysicalComplexity.SAT3HardFamily
PH22 DecisionQuotient.PhysicalComplexity.p_eq_np_physically_impossible_via_sat_hard_family
PH23 DecisionQuotient.PhysicalComplexity.collapse_possible_without_positive_bit_cost
PH24 DecisionQuotient.PhysicalComplexity.collapse_possible_without_exponential_lower_bound
PH25 DecisionQuotient.PhysicalComplexity.no_go_transfer_requires_collapse_map
PH26 DecisionQuotient.PhysicalComplexity.no_collapse_of_bounded_budget_pos_cost_exp_lb
PH27 DecisionQuotient.PhysicalComplexity.collapse_implies_assumption_failure_disjunction
PH28 DecisionQuotient.PhysicalComplexity.deterministic_no_physical_collapse
PH29 DecisionQuotient.PhysicalComplexity.probabilistic_no_physical_collapse
PH30 DecisionQuotient.PhysicalComplexity.sequential_no_physical_collapse
PH31 DecisionQuotient.PhysicalComplexity.collapse_possible_with_unbounded_budget_profile
PH32 DecisionQuotient.PhysicalComplexity.exp_budget_profile_unbounded
PH33 DecisionQuotient.PhysicalComplexity.finite_budget_assumption_is_necessary
PH34 DecisionQuotient.PhysicalComplexity.CoherentDQRejectionAtRequirement
PH35 DecisionQuotient.PhysicalComplexity.coherent_dq_rejection_impossible_at_requirement
PH36 DecisionQuotient.PhysicalComplexity.coherent_dq_rejection_impossible_canonical
PI1 DecisionQuotient.Physics.PhysicalIncompleteness.UniverseModel
PI2 DecisionQuotient.Physics.PhysicalIncompleteness.PhysicallyInstantiated
PI3 DecisionQuotient.Physics.PhysicalIncompleteness.no_surjective_instantiation_of_card_gap
PI4 DecisionQuotient.Physics.PhysicalIncompleteness.physical_incompleteness_of_card_gap
PI5 DecisionQuotient.Physics.PhysicalIncompleteness.physical_incompleteness_of_bounds
PI6 DecisionQuotient.Physics.PhysicalIncompleteness.under_resolution_implies_collision
PI7 DecisionQuotient.Physics.PhysicalIncompleteness.under_resolution_implies_decision_collision
PS1 DecisionQuotient.Physics.ClaimTransport.PhysicalStateSemantics
PS2 DecisionQuotient.Physics.ClaimTransport.physical_state_has_witness
PS3 DecisionQuotient.Physics.ClaimTransport.physical_state_claim_of_instance_claim
PS4 DecisionQuotient.Physics.ClaimTransport.physical_state_claim_of_universal_core
PYH1 Python.python_has_hooks
PYI1 Python.python_has_introspection
QT1 DecisionQuotient.DecisionProblem.quotient_is_coarsest
QT2 DecisionQuotient.DecisionProblem.quotientMap_preservesOpt
QT3 DecisionQuotient.DecisionProblem.quotient_represents_opt_equiv
QT4 DecisionQuotient.DecisionProblem.factors_implies_respects
QT6 DecisionQuotient.DecisionProblem.quotientEquivOptRange_apply_quotientMap
QT7 DecisionQuotient.DecisionProblem.quotient_has_unique_factorization
RAT1 ClaimClosure.rate_incoherence_step
RD1 DecisionQuotient.Information.shannonEntropy_nonneg
RD2 DecisionQuotient.Information.rate_zero_distortion
RD3 DecisionQuotient.Information.rate_monotone

Lean handle entry (continued)
RED1 ClaimClosure.redundancy_incoherence_equiv
REG1 ClaimClosure.operating_regimes_partition
REG2 ClaimClosure.pareto_optimality_p1
REG3 ClaimClosure.no_tradeoff_at_p1
REG4 ClaimClosure.amortized_complexity_core
REQ1 structural_facts_fixed_at_definition
REQ2 definition_hooks_necessary
REQ3 introspection_necessary_for_verification
RS1 DecisionQuotient.Information.equiv_preserves_decision
RS2 DecisionQuotient.Information.rate_equals_srank
RS3 DecisionQuotient.Information.compression_below_srank_fails
RS4 DecisionQuotient.Information.srank_bits_sufficient
RS5 DecisionQuotient.Information.rate_distortion_bridge
S2P1 DecisionQuotient.Sigma2PHardness.exactlyIdentifiesRelevant_iff_sufficient_and_subset_relevantFinset
S2P2 DecisionQuotient.Sigma2PHardness.min_representationGap_zero_iff_relevant_card
S2P3 DecisionQuotient.Sigma2PHardness.min_sufficient_set_iff_relevant_card
S2P4 DecisionQuotient.Sigma2PHardness.representationGap
S2P5 DecisionQuotient.Sigma2PHardness.representationGap_eq_waste_plus_missing
S2P6 DecisionQuotient.Sigma2PHardness.representationGap_eq_zero_iff
S2P7 DecisionQuotient.Sigma2PHardness.representationGap_missing_eq_gapCard
S2P8 DecisionQuotient.Sigma2PHardness.representationGap_zero_iff_minimalSufficient
S2P9 DecisionQuotient.Sigma2PHardness.sufficient_iff_relevant_subset
SE1 DecisionQuotient.ClaimClosure.SE1
SE2 DecisionQuotient.ClaimClosure.SE2
SE3 DecisionQuotient.ClaimClosure.SE3
SE4 DecisionQuotient.ClaimClosure.SE4
SE5 DecisionQuotient.ClaimClosure.SE5
SE6 DecisionQuotient.ClaimClosure.SE6
SID1 ClaimClosure.dof1_zero_side_information
SID2 ClaimClosure.side_information_scales_with_redundancy
SK1 DecisionQuotient.DecisionProblem.srank_eq_relevant_card
SK2 DecisionQuotient.DecisionProblem.srank_le_n
SK3 DecisionQuotient.DecisionProblem.srank_zero_iff_constant
SOT1 ssot_iff
SR1 DecisionQuotient.ClaimClosure.SR1
SR2 DecisionQuotient.ClaimClosure.SR2
SR3 DecisionQuotient.ClaimClosure.SR3
SR4 DecisionQuotient.ClaimClosure.SR4
SR5 DecisionQuotient.ClaimClosure.SR5
TC1 DecisionQuotient.ToolCollapse.WorkProfile
TC2 DecisionQuotient.ToolCollapse.WorkModel
TC3 DecisionQuotient.ToolCollapse.ToolModel
TC4 DecisionQuotient.ToolCollapse.EventualStrictImprovement

Lean handle entry (continued)
TC5 DecisionQuotient.ToolCollapse.EffectiveCollapse
TC6 DecisionQuotient.ToolCollapse.tool_never_worse
TC7 DecisionQuotient.ToolCollapse.strict_improvement_has_witness
TC8 DecisionQuotient.ToolCollapse.effective_collapse_of_eventual_strict
TC9 DecisionQuotient.ToolCollapse.expBaseline
TC10 DecisionQuotient.ToolCollapse.linearTool
TC11 DecisionQuotient.ToolCollapse.linear_tool_eventual_strict
TC12 DecisionQuotient.ToolCollapse.linear_tool_effective_collapse
TL3 DecisionQuotient.ThermodynamicLift.joulesPerBit_pos_of_landauer_floor
TL4 DecisionQuotient.ThermodynamicLift.energy_lower_mandatory_of_landauer_floor
TL5 DecisionQuotient.ThermodynamicLift.joulesPerBit_pos_of_landauer_calibration
TL6 DecisionQuotient.ThermodynamicLift.energy_lower_mandatory_of_landauer_calibration
TRI1 timing_trichotomy_exhaustive
TRI2 trichotomy_necessary_for_causality
TRI3 trichotomy_necessary_for_mechanism
TRI4 no_mechanism_outside_trichotomy
TUR1 DecisionQuotient.Physics.transitionProb_nonneg
TUR2 DecisionQuotient.Physics.transitionProb_sum_one
TUR5 DecisionQuotient.Physics.tur_bridge
TUR6 DecisionQuotient.Physics.multiple_futures_entropy_production
UNIQ1 unique_tree_encoding
UNQ1 uniqueness
UNQ2 uniqueness_exists
UNQ3 uniqueness_unique
UO1 DecisionQuotient.UniverseDynamics
UO2 DecisionQuotient.feasibleActions
UO3 DecisionQuotient.lawDecisionProblem
UO4 DecisionQuotient.lawUtility
UO5 DecisionQuotient.logicallyDeterministic
UO6 DecisionQuotient.universe_sets_objective_schema
UO7 DecisionQuotient.lawUtility_eq_of_allowed_iff
UO8 DecisionQuotient.opt_eq_feasible_of_gap
UO9 DecisionQuotient.infeasible_not_optimal_of_gap
UO10 DecisionQuotient.opt_singleton_of_logicallyDeterministic
UO11 DecisionQuotient.opt_eq_of_allowed_iff
UQ1 DecisionQuotient.Physics.Uncertainty.binaryIdentityProblem
UQ2 DecisionQuotient.Physics.Uncertainty.binaryIdentityProblem_opt_true
UQ3 DecisionQuotient.Physics.Uncertainty.binaryIdentityProblem_opt_false
UQ4 DecisionQuotient.Physics.Uncertainty.exists_decision_problem_with_nontrivial_opt
UQ5 DecisionQuotient.Physics.Uncertainty.PhysicalNontrivialOptAssumption
UQ6 DecisionQuotient.Physics.Uncertainty.exists_decision_problem_with_nontrivial_opt_of_physical
W1 DecisionQuotient.Physics.single_future_zero_cost
W2 DecisionQuotient.Physics.transportCost_pos_of_offDiag

Lean handle entry (continued)

W3 DecisionQuotient.Physics.integrity_is_centroid

W4 DecisionQuotient.Physics.wasserstein_bridge

WC1 DecisionQuotient.Physics.WolpertConstraints.landauer_floor_plus_overhead_lower_bound

WC2 DecisionQuotient.Physics.WolpertConstraints.effective_model_dominates_landauer_floor

WC3 DecisionQuotient.Physics.WolpertConstraints.effective_model_strictly_exceeds_landauer_of_strict_overhead

WC4 DecisionQuotient.Physics.WolpertConstraints.energy_lower_bound_mono_under_overhead

WC5 DecisionQuotient.Physics.WolpertConstraints.physical_grounding_bundle_with_wolpert_overhead

WD1 DecisionQuotient.checking_witnessing_duality_budget

WD2 DecisionQuotient.no_sound_checker_below_witness_budget

WD3 DecisionQuotient.checking_time_ge_witness_budget

WD4 DecisionQuotient.witnessBudgetEmpty

WD5 DecisionQuotient.checkingBudgetPairs

WM1 DecisionQuotient.Physics.WolpertMismatch.mismatchKL_nonneg

WM2 DecisionQuotient.Physics.WolpertMismatch.mismatchKL_eq_zero_iff_eq

WM3 DecisionQuotient.Physics.WolpertMismatch.mismatchKL_pos_of_exists_ne

WM4 DecisionQuotient.Physics.WolpertMismatch.mismatchNatLowerBound_pos_of_exists_ne

WM5 DecisionQuotient.Physics.WolpertDecomposition.periodic_modular_mismatch_of_distribution_mismatch

WM6
DecisionQuotient.Physics.WolpertDecomposition.effective_model_strictly_exceeds_landauer_of_distribution_mismatch

WP1 DecisionQuotient.Physics.WolpertDecomposition.DecomposedProcessModel.totalOverheadPerBit_eq_sum

WP2 DecisionQuotient.Physics.WolpertDecomposition.landauer_floor_plus_decomposition_lower_bound

WP3 DecisionQuotient.Physics.WolpertDecomposition.effective_model_dominates_landauer_floor_decomposition

WP4 DecisionQuotient.Physics.WolpertDecomposition.effective_model_strictly_exceeds_landauer_of_periodic_modular_
mismatch

WP5
DecisionQuotient.Physics.WolpertDecomposition.effective_model_strictly_exceeds_landauer_of_stopping_time_residual

WP6
DecisionQuotient.Physics.WolpertDecomposition.effective_model_strictly_exceeds_landauer_of_either_cited_component

WP7 DecisionQuotient.Physics.WolpertDecomposition.landauer_floor_plus_structural_resource_lower_bound

WP8 DecisionQuotient.Physics.WolpertDecomposition.energy_lower_bound_increases_by_structural_resource

WP9 DecisionQuotient.Physics.WolpertDecomposition.physical_grounding_bundle_with_wolpert_decomposition

WR1 DecisionQuotient.Physics.WolpertResidual.pairwiseResidualKL_nonneg

WR2 DecisionQuotient.Physics.WolpertResidual.pairwiseResidualKL_pos_of_asymmetry

WR3 DecisionQuotient.Physics.WolpertResidual.residualNatLowerBound_pos_of_asymmetry

WR4 DecisionQuotient.Physics.WolpertDecomposition.stopping_time_residual_of_pairwise_flow_asymmetry

WR5
DecisionQuotient.Physics.WolpertDecomposition.effective_model_strictly_exceeds_landauer_of_pairwise_flow_asymmetry

WR6 DecisionQuotient.Physics.WolpertResidual.discreteResidualNatLowerBound_pos_of_asymmetry_or_oneway

WR7 DecisionQuotient.Physics.WolpertDecomposition.stopping_time_residual_of_discrete_edge_split

WR8 DecisionQuotient.Physics.WolpertDecomposition.effective_model_strictly_exceeds_landauer_of_discrete_edge_split

WR9 DecisionQuotient.Physics.WolpertDecomposition.stopping_time_residual_of_finite_discrete_witness

WR10
DecisionQuotient.Physics.WolpertDecomposition.effective_model_strictly_exceeds_landauer_of_finite_discrete_witness

XC1 DecisionQuotient.Physics.srank_determines_states

XC2 DecisionQuotient.Physics.more_states_more_transport

Lean handle entry (continued)
XC3 DecisionQuotient.Physics.transport_lower_bound
XC4 DecisionQuotient.Physics.transport_independent_of_energy
XC5 DecisionQuotient.Physics.transport_independent_of_precision
XC6 DecisionQuotient.Physics.srank_unified_complexity
XC7 DecisionQuotient.Physics.complete_bridge_set

A.4 Proof Hardness Index

Paper handle	Hardness profile	Regime tags	Lean support
cor:dof-errors	unspecified	-	L15, L23
cor:dof-monotone	unspecified	-	L33
cor:dof-ratio	unspecified	-	L21
cor:leverage-energy	unspecified	-	L3, L6
cor:linear-approx	unspecified	-	L16, L39
cor:physical-assumption-necessity	unspecified	-	L7, L8
prop:dof-additive	unspecified	-	L17, L19
thm:approx-bound	unspecified	-	L32, L39
thm:composition	unspecified	-	L18, L19
thm:dof-reliability	unspecified	-	L22, L27
thm:england	unspecified	-	L45
thm:error-compound	unspecified	-	L20, L41
thm: error-independence	unspecified	-	L23
thm:error-prob	unspecified	-	L24, L40
thm:expected-errors	unspecified	-	L25, L26
thm:five-way	unspecified	-	*(no derived Lean handle found)*
thm:leverage-error	unspecified	-	L29
thm:leverage-gap	unspecified	-	L30
thm:leverage-max	unspecified	-	L28, L31
thm:metaprog	unspecified	-	L35, L36
thm:mod-bound	unspecified	-	L37
thm:nominal-leverage	unspecified	-	L12, L13
thm:optimal	unspecified	-	L34, L38
thm: paper1-integration	unspecified	-	L13, L14
thm: paper2-integration	unspecified	-	L10, L11
thm:physical-budget-boundary	unspecified	-	L2, L4
thm:physical-energy-floor	unspecified	-	L1, L5
thm:ssot-leverage	unspecified	-	L9, L11
thm: testable-prediction	unspecified	-	L42

Auto summary: indexed 29 claims by hardness profile (unspecified=29).

B Notes on assumptions and extensions

This appendix lists the principal modeling assumptions and common extensions relevant when applying the leverage framework:

- **Independence and orthogonality:** Error-independence at the DOF level is derived from axis orthogonality assumptions; when dependencies are present, leverage remains a useful comparative metric but probabilistic models should be adjusted to account for correlation.

- **Capability counting:** The core theorems require only relative ordering of capabilities; cardinality serves as a practical proxy for capability breadth in examples and case studies.
- **Multi-objective concerns:** Leverage addresses error probability specifically. Performance, security, and other attributes require multi-objective analysis (e.g., Pareto frontiers) before operational decisions.
- **Implementations:** SSOT and other instantiations depend on language and platform features; the theoretical principle is implementation-agnostic.
- **Future work:** Extensions include explicit correlated-error models, weighted capability measures, and integration into multi-criteria architectural decision frameworks.

C Complete Theorem Index

Paper-level labeled claims in this manuscript:

Foundations (Section 2):

- Proposition 2.4 (DOF Additivity)
- Theorem 2.12 (Modification Bounded by DOF)

Probability Model (Section 3):

- Theorem 3.1
- Corollary 3.2
- Theorem 3.3
- Theorem 3.4
- Corollary 3.5
- Corollary 3.6
- Theorem 3.7
- Theorem 3.9
- Theorem 3.10
- Theorem 3.11
- Theorem 3.12
- Corollary 3.13

Main Results (Section 4):

- Theorem 4.1
- Theorem 4.2
- Theorem 4.3

- Corollary 4.4
- Theorem 4.5
- Corollary 4.6
- Theorem 4.7
- Theorem 4.8
- Theorem 4.9
- Theorem 4.11
- Theorem 4.12

Instances (Section 5):

- Theorem 6.3
- Theorem 6.7

Mechanization status: 48509 lines, 2079 theorems/lemmas, 0 `sorry`, 195 files.
Primary Lean sources:

- `Leverage/Foundations.lean`
- `Leverage/Probability.lean`
- `Leverage/Theorems.lean`
- `Leverage/Physical.lean`
- `Leverage/SSOT.lean`
- `Leverage/Typing.lean`
- `Leverage/Examples.lean`
- `Leverage/WeightedLeverage.lean`
- `LambdaDR.lean`
- `Leverage.lean`

References

[1] Robert Allen and David Garlan. A formal basis for architectural connection. *ACM Transactions on Software Engineering and Methodology (TOSEM)*, 6(3):213–249, 1997.

[2] Felix Bachmann, Len Bass, and Constance Heitmeyer. Characterizing, planning, and managing technical debt. Technical report, Software Engineering Institute, 2000.

[3] Victor R. Basili, Lionel C. Briand, and Walcélio L. Melo. A validation of object-oriented design metrics as quality indicators. *IEEE Transactions on Software Engineering*, 22(10):751–761, 1996.

[4] Len Bass, Paul Clements, and Rick Kazman. *Software Architecture in Practice*. Addison-Wesley Professional, 2012.

[5] Daniel G. Bobrow, Kenneth Kahn, Gregor Kiczales, Larry Masinter, Mark Stefik, and Frank Zdybel. Commonloops: Merging lisp and object-oriented programming. *ACM SIGPLAN Notices*, 21(11):17–29, 1986.

[6] Shyam R. Chidamber and Chris F. Kemerer. A metrics suite for object oriented design. *IEEE Transactions on Software Engineering*, 20(6):476–493, 1994.

[7] Edgar F. Codd. A relational model of data for large shared data banks. *Communications of the ACM*, 13(6):377–387, 1970.

[8] Ward Cunningham. The wycash portfolio management system. In *OOPSLA '92 Addendum to the Proceedings*, pages 29–30. ACM, 1992.

[9] Christopher J. Date. *An Introduction to Database Systems*. Pearson Education, 2003.

[10] Martin Fowler and James Lewis. Microservices: a definition of this new architectural term. https://martinfowler.com/articles/microservices.html, 2014.

[11] Erich Gamma, Richard Helm, Ralph Johnson, and John Vlissides. *Design Patterns: Elements of Reusable Object-Oriented Software*. Addison-Wesley Professional, 1994.

[12] David Garlan, Robert Allen, and John Ockerbloom. Exploiting style in architectural design environments. In *Proceedings of the 2nd ACM SIGSOFT Symposium on Foundations of Software Engineering*, pages 175–188, 1994.

[13] David Garlan, Robert T. Monroe, and David Wile. Acme: An architecture description interchange language. In *Proceedings of CASCON '97*, pages 169–183, 1997.

[14] David Heinemeier Hansson. The rails doctrine. https://rubyonrails.org/doctrine, 2005.

[15] Rick Kazman, Mark Klein, and Paul Clements. Atam: Method for architecture evaluation. Technical Report CMU/SEI-2000-TR-004, Carnegie Mellon University, Software Engineering Institute, 2000.

[16] Gregor Kiczales, Jim des Rivières, and Daniel G. Bobrow. *The Art of the Metaobject Protocol*. MIT Press, 1991.

[17] Julian Mackay, Sophia Drossopoulou, James Noble, and Susan Eisenbach. Necessity specifications for robustness. Zenodo, 2022. Coq mechanization of necessity specifications for module robustness.

[18] Pattie Maes. Concepts and experiments in computational reflection. *ACM SIGPLAN Notices*, 22(12):147–155, 1987.

[19] Thomas J. McCabe. A complexity measure. *IEEE Transactions on Software Engineering*, SE-2(4):308–320, 1976.

[20] David A. Patterson and John L. Hennessy. *Computer Organization and Design: The Hardware/Software Interface*. Morgan Kaufmann, 2013. Discusses RAID and reliability formulas.

[21] Rob Pike and Brian W. Kernighan. Program design in the unix environment. *AT&T Bell Laboratories Technical Journal*, 63(8):1595–1605, 1984.

[22] Mary Shaw and David Garlan. *Software Architecture: Perspectives on an Emerging Discipline*. Prentice Hall, 1996.

[23] Tristan Simas. Computational complexity of physical counting. Zenodo, 2025. Companion paper. Proves coNP-hardness of sufficiency checking, tractable subcases, and P $\neq$ NP from physical axioms (Landauer, finite energy, finite signal speed, nontrivial state space). All results machine-checked in Lean 4.

[24] Tristan Simas. Identification capacity and rate-query tradeoffs in classification systems. Zenodo, 2025. Paper 1 of this series. Proves axis orthogonality theorem.

[25] Tristan Simas. Zero-incoherence capacity of interactive encoding systems: Achievability, converse, and side information bounds. Zenodo, 2025. Paper 2 of this series. Proves coherence impossibility theorems for programming languages.

[26] Wayne P. Stevens, Glenford J. Myers, and Larry L. Constantine. Structured design. *IBM Systems Journal*, 13(2):115–139, 1974.